

Nonparametric Bayesian Methods



Tamara Broderick

ITT Career Development Assistant Professor
EECS
MIT

Michael I. Jordan

Pehong Chen Distinguished Professor
EECS, Statistics
UC Berkeley

Nonparametric Bayesian Methods: Part I

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Nonparametric Bayes

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- Bayesian

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$$\mathbb{P}(\text{parameters}|\text{data}) \propto \mathbb{P}(\text{data}|\text{parameters})\mathbb{P}(\text{parameters})$$

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- Not parametric (i.e. not finite parameter, unbounded/
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[wikipedia.org]

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[wikipedia.org]

“Wikipedia phenomenon”

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Nonparametric Bayes

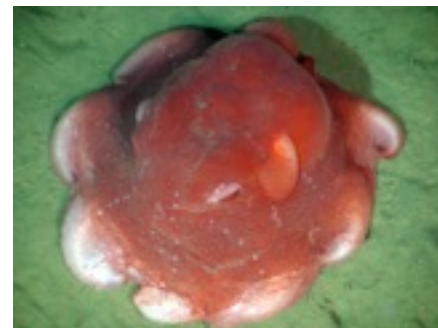
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[Ed Bowlby, NOAA]

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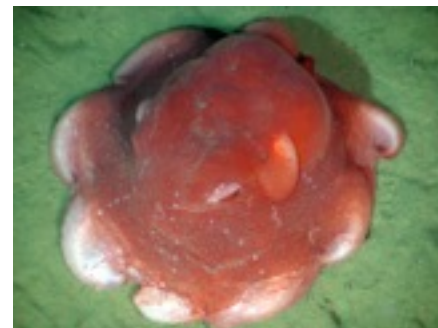
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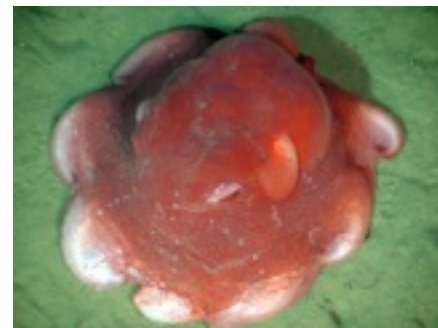
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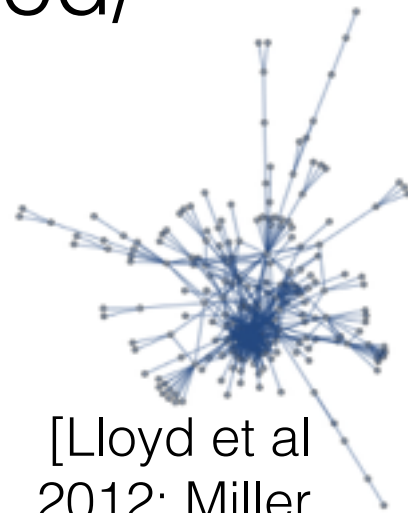
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[Lloyd et al 2012; Miller et al 2010]



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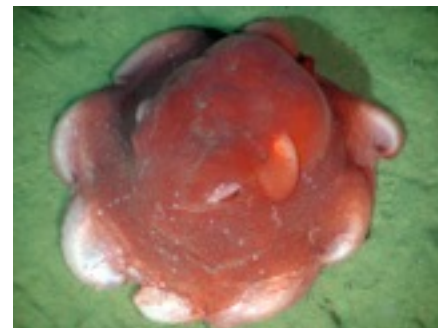
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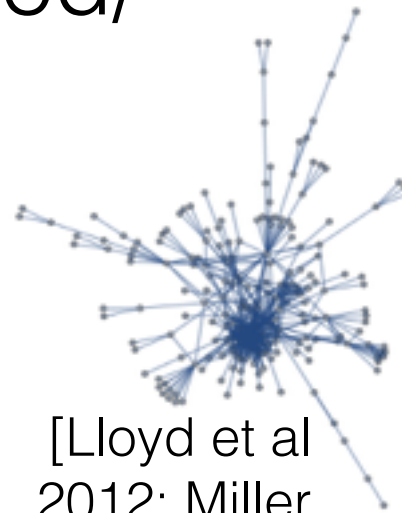
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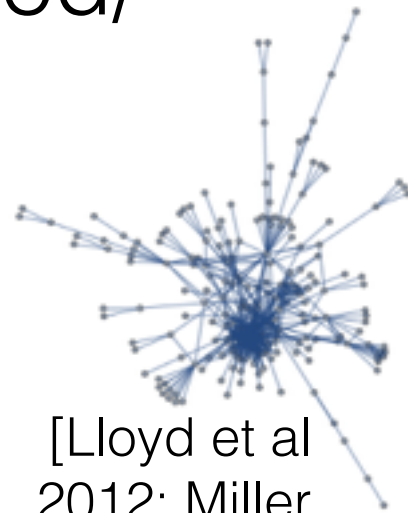
[Sudderth, Jordan 2009]

Nonparametric Bayes

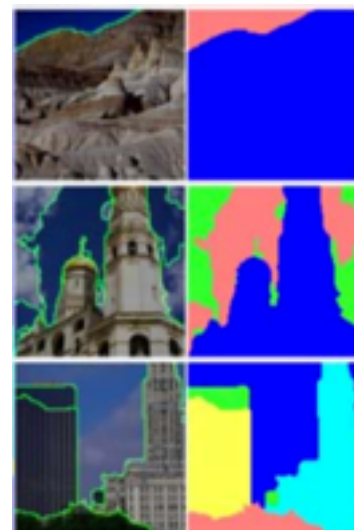
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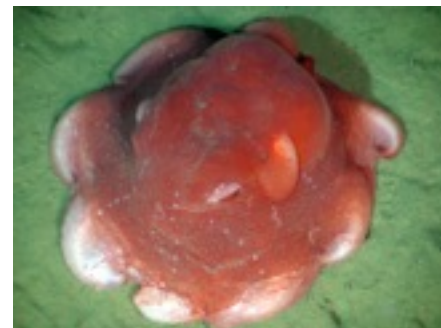
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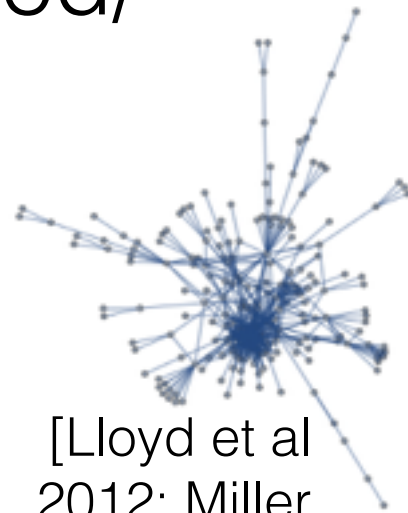
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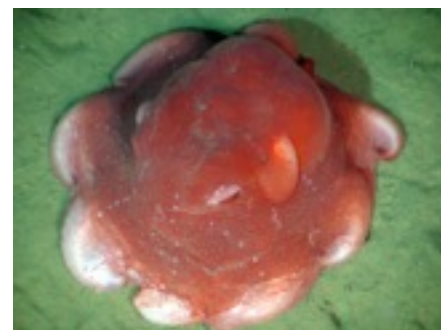
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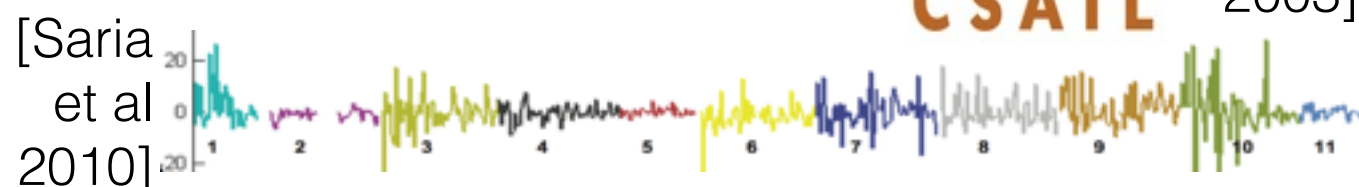
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[wikipedia.org]

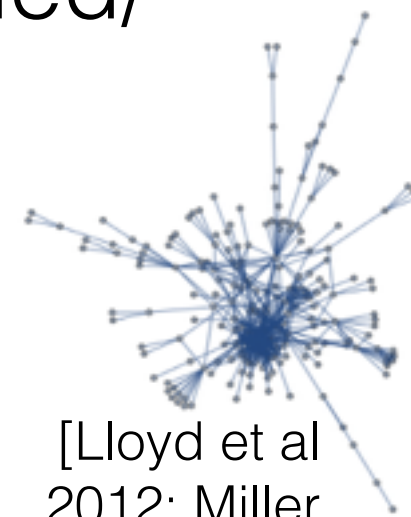


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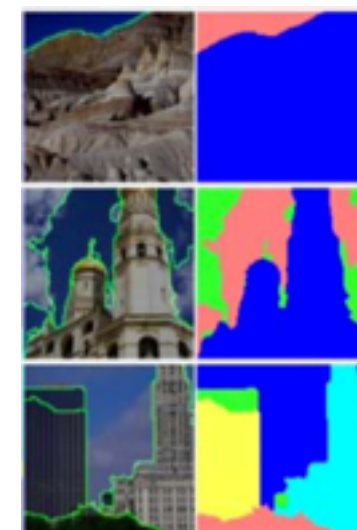
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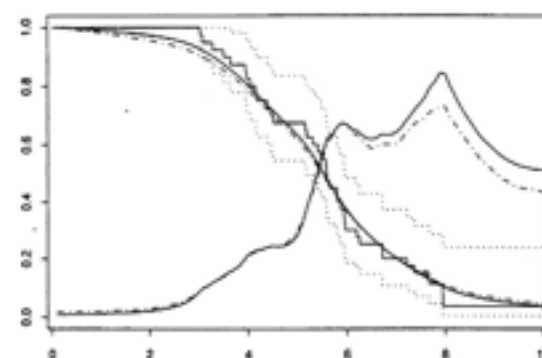
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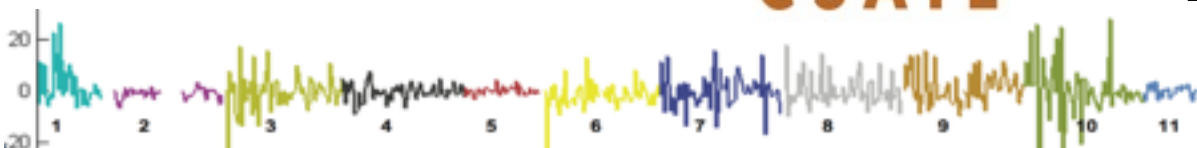


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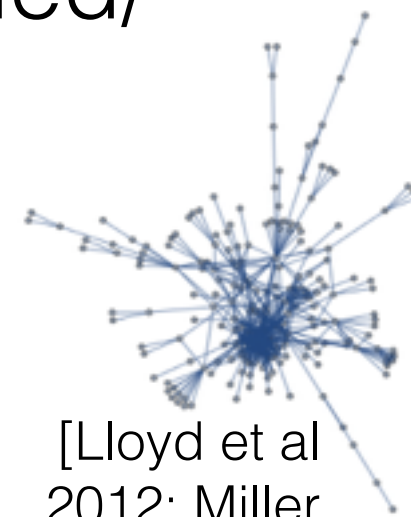


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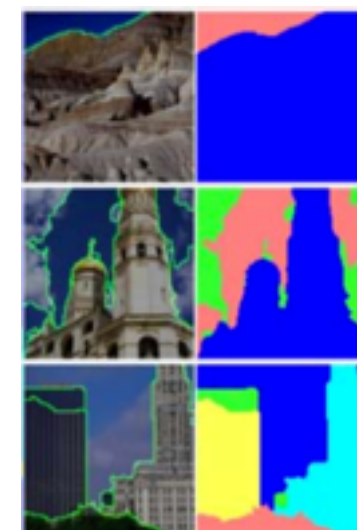
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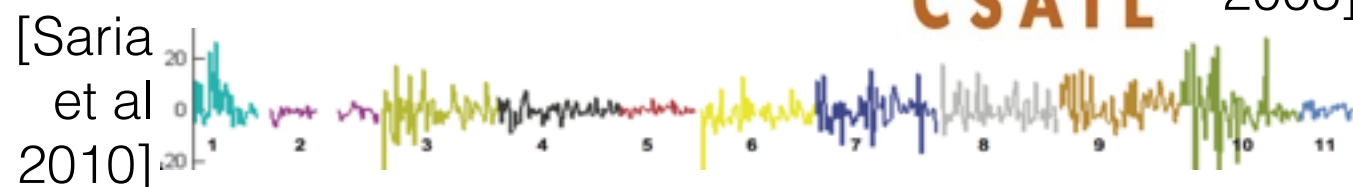
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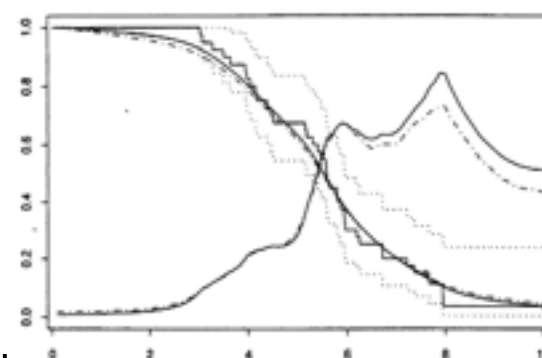
[Ewens 1972; Hartl, Clark 2003]



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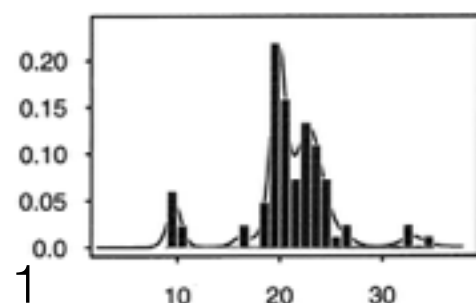
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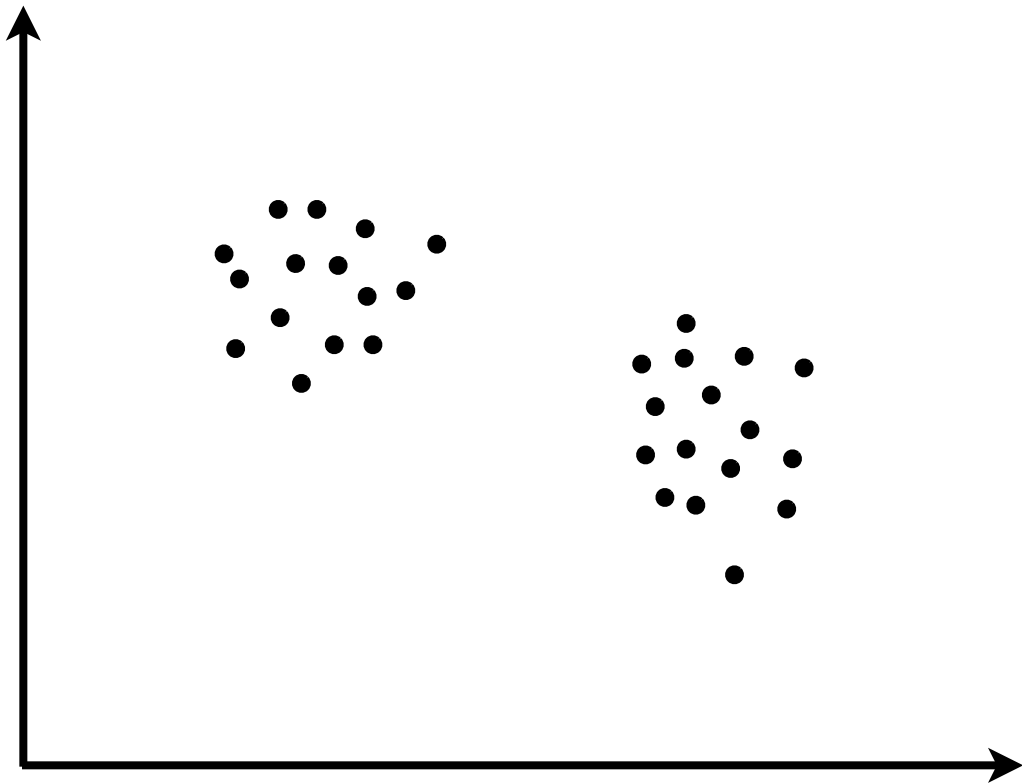


[Escobar, West 1995; Ghosal et al 1999]

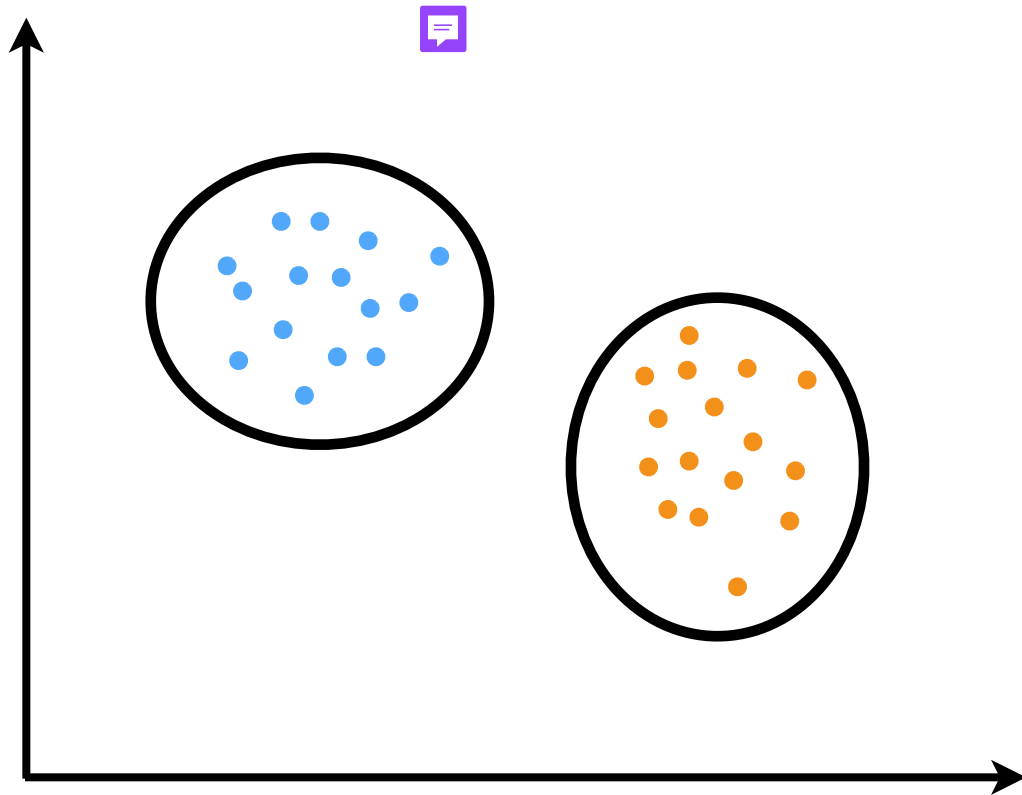
Roadmap

- Example problem: clustering
- Example NPBayes model: Dirichlet process
- Big questions
 - Why NPBayes?
 - What does a growing/infinite number of parameters really mean (in NPBayes)? 
 - Why is NPBayes challenging but practical?

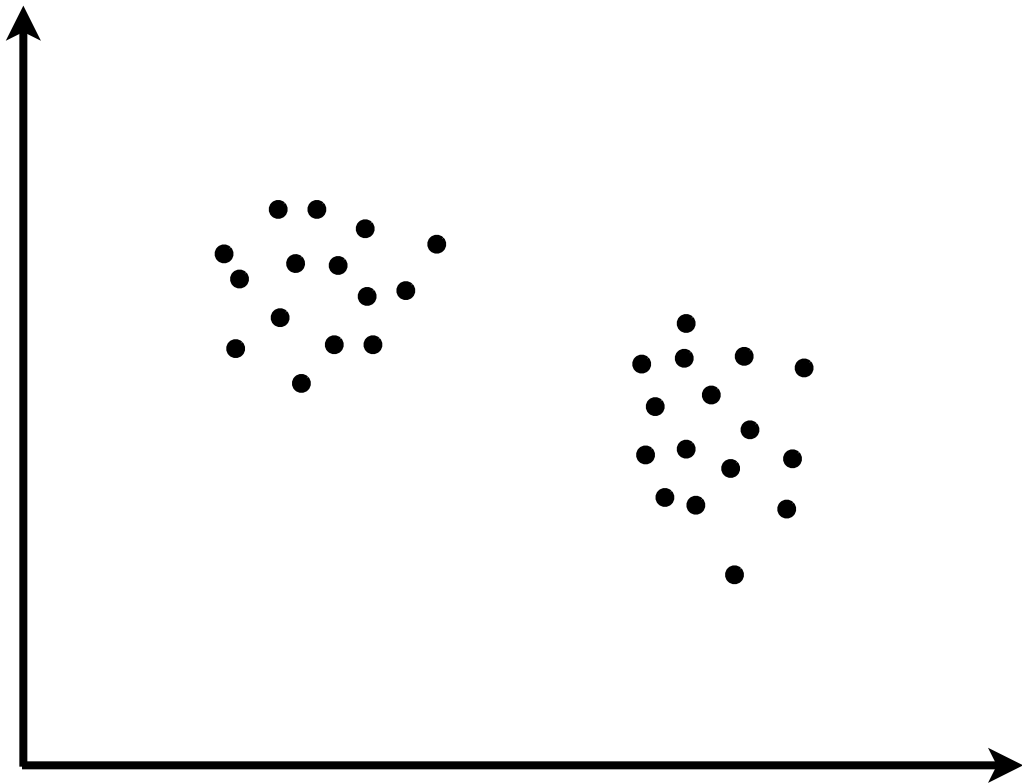
Clustering



Clustering



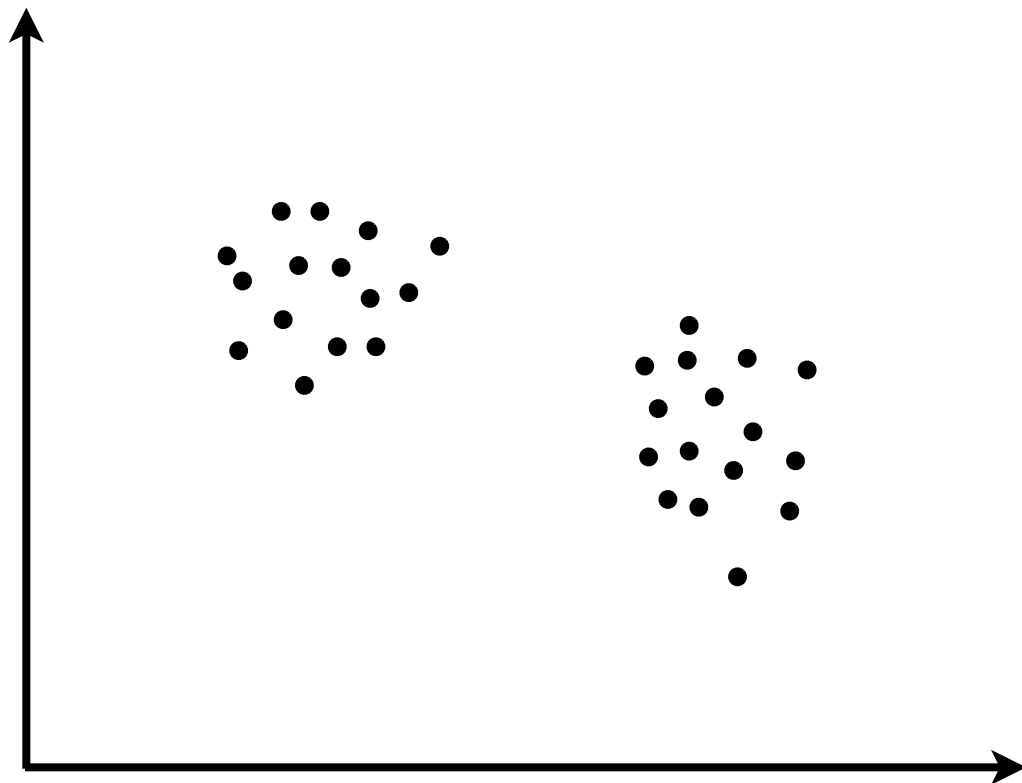
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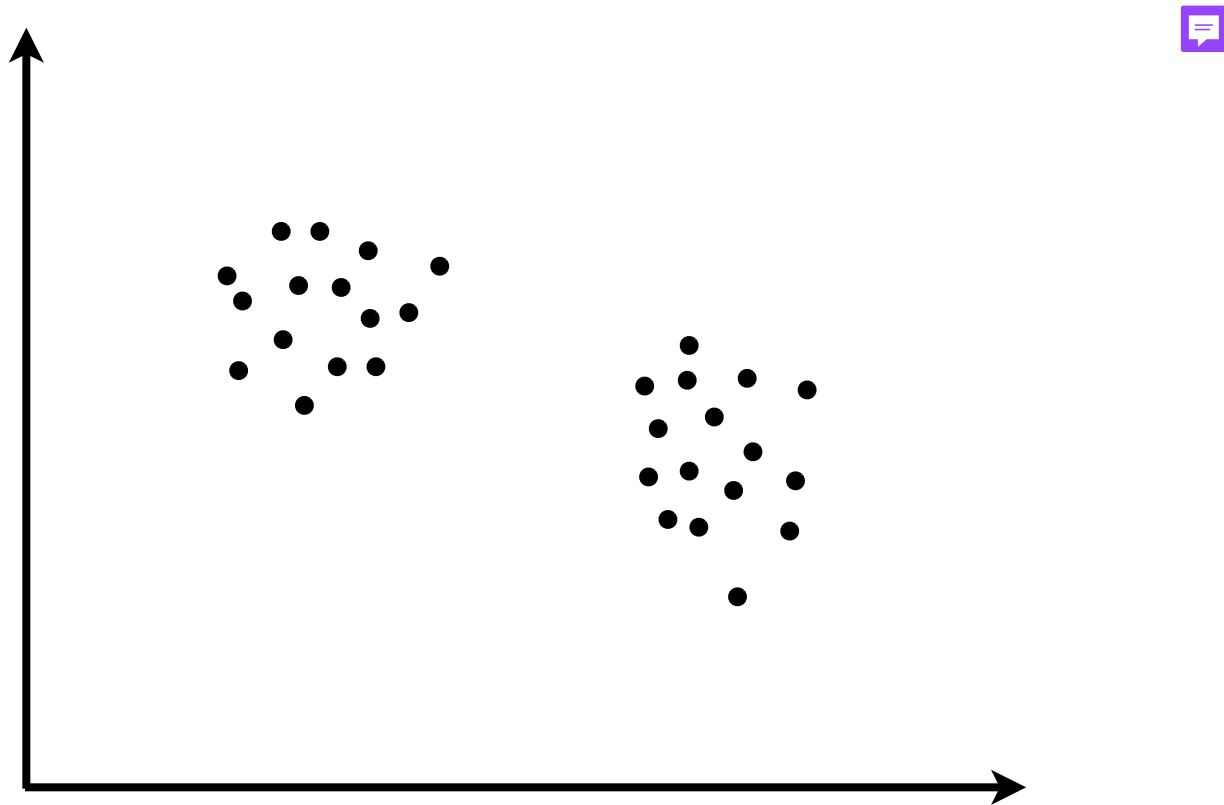


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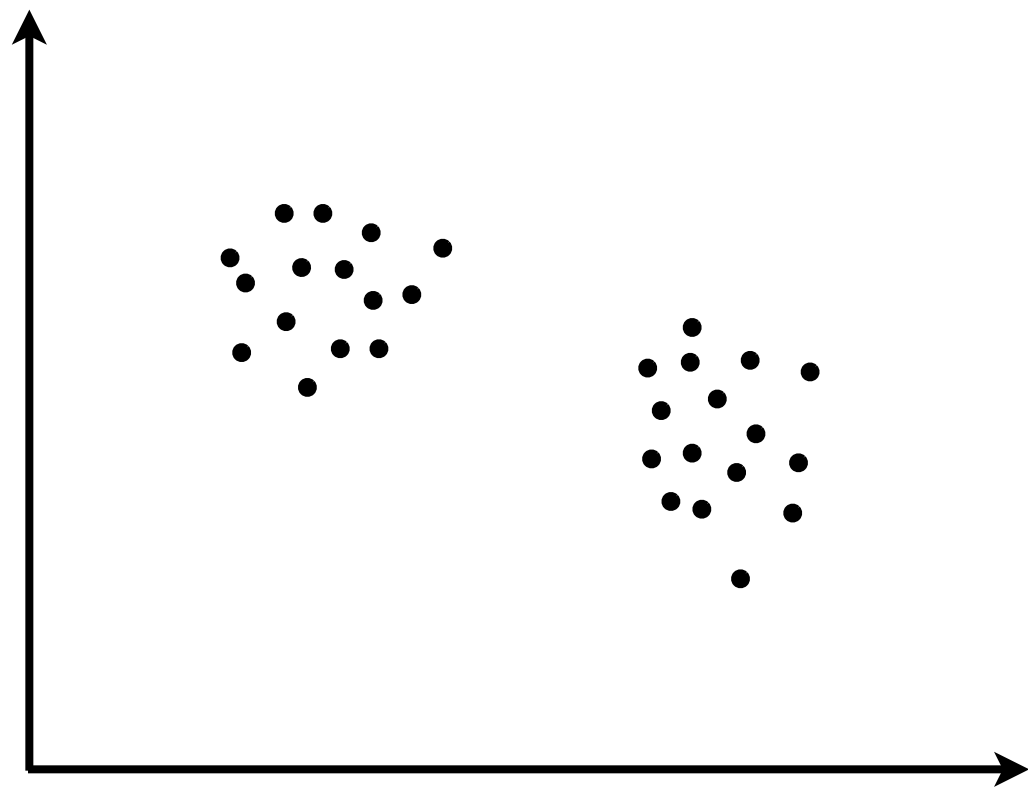
Generative model

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Generative model

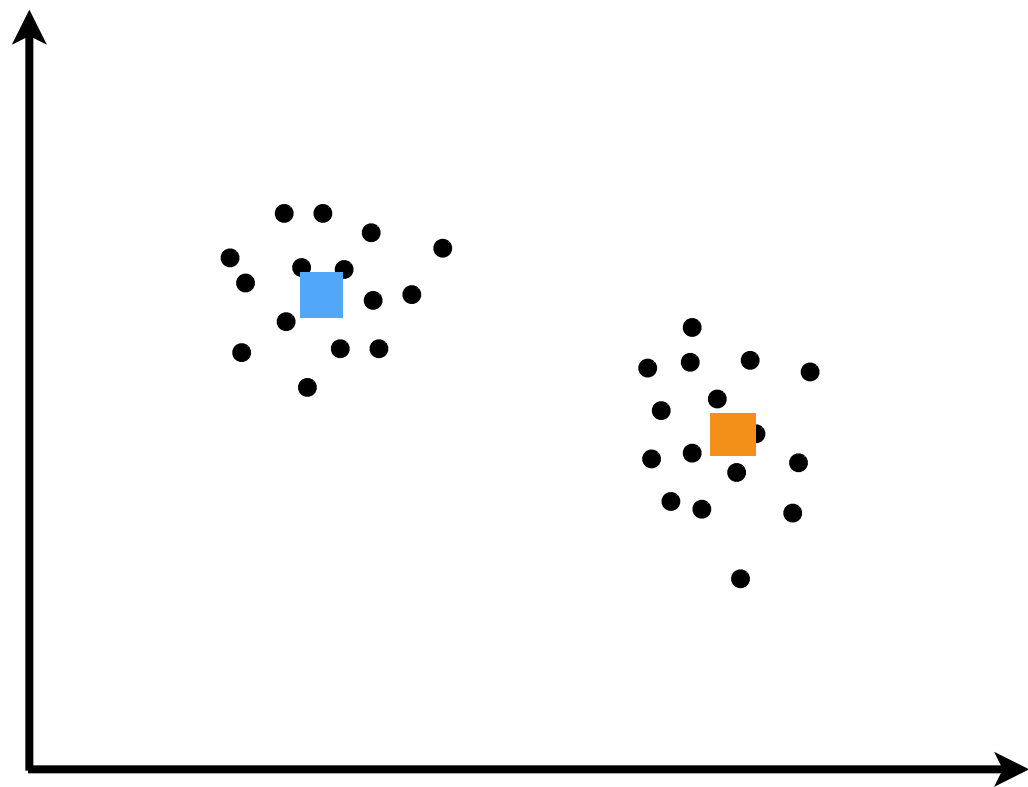
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- Finite Gaussian mixture model ($K=2$ clusters)

Generative model

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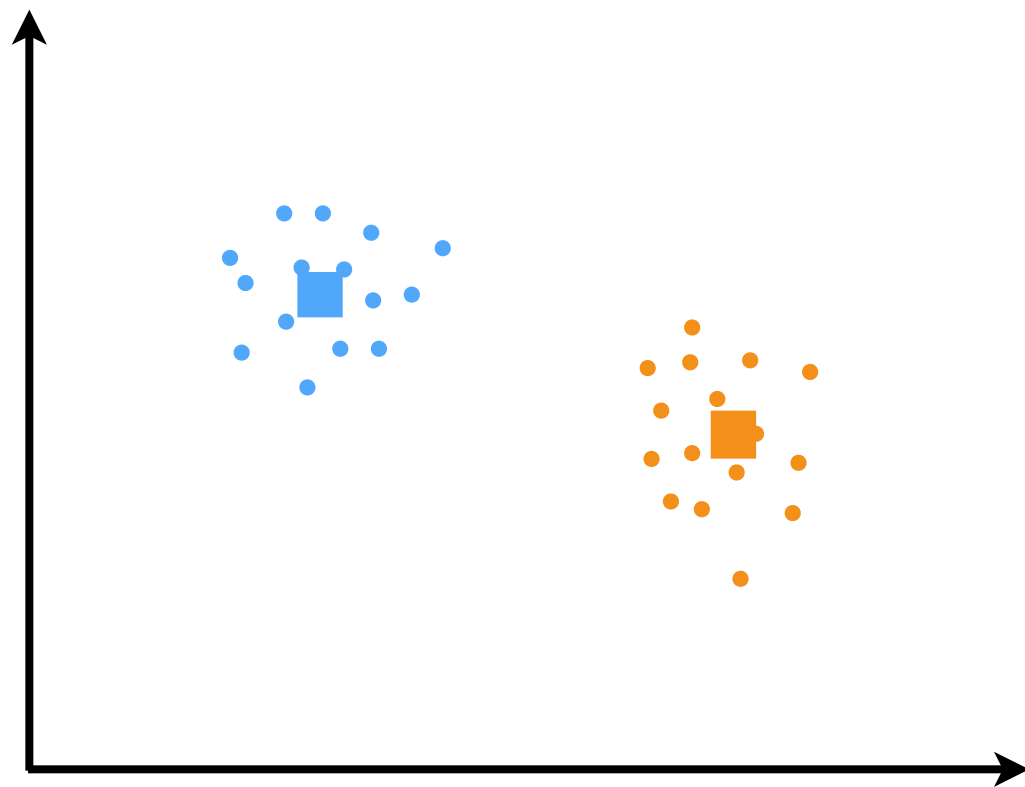
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μ_k

Generative model

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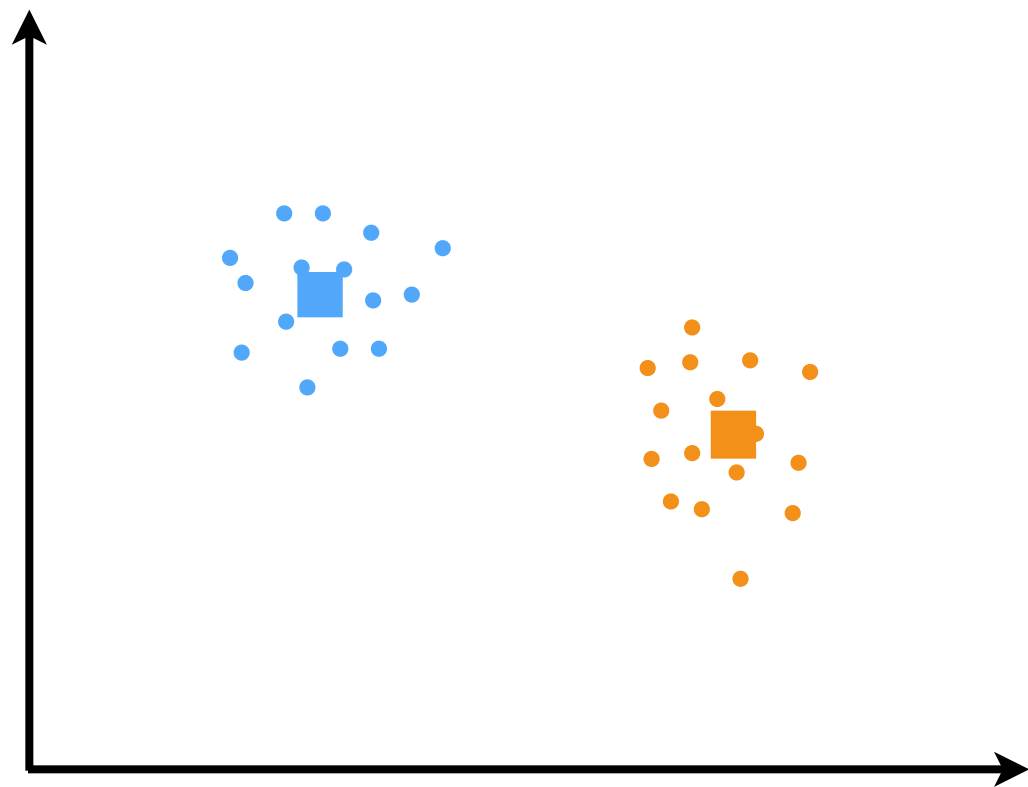
μ_k



z_n

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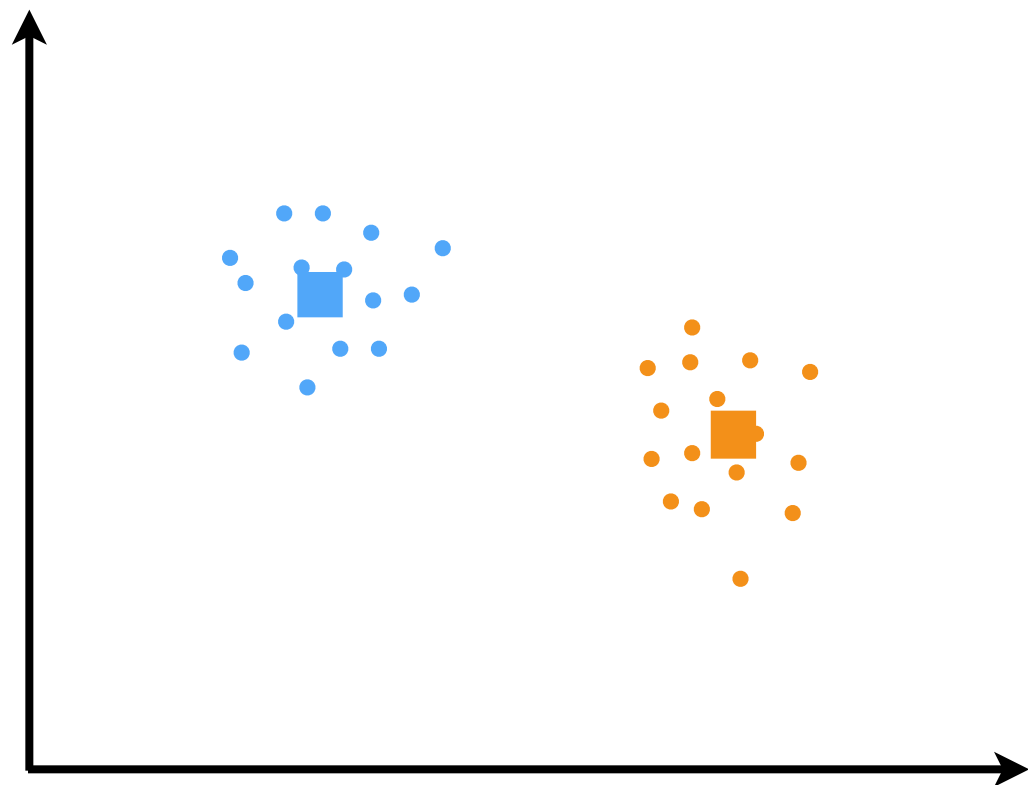
z_n

$x_n \stackrel{\text{indep}}{\sim} \mathcal{N}(\mu_{z_n}, \Sigma)$

Generative model



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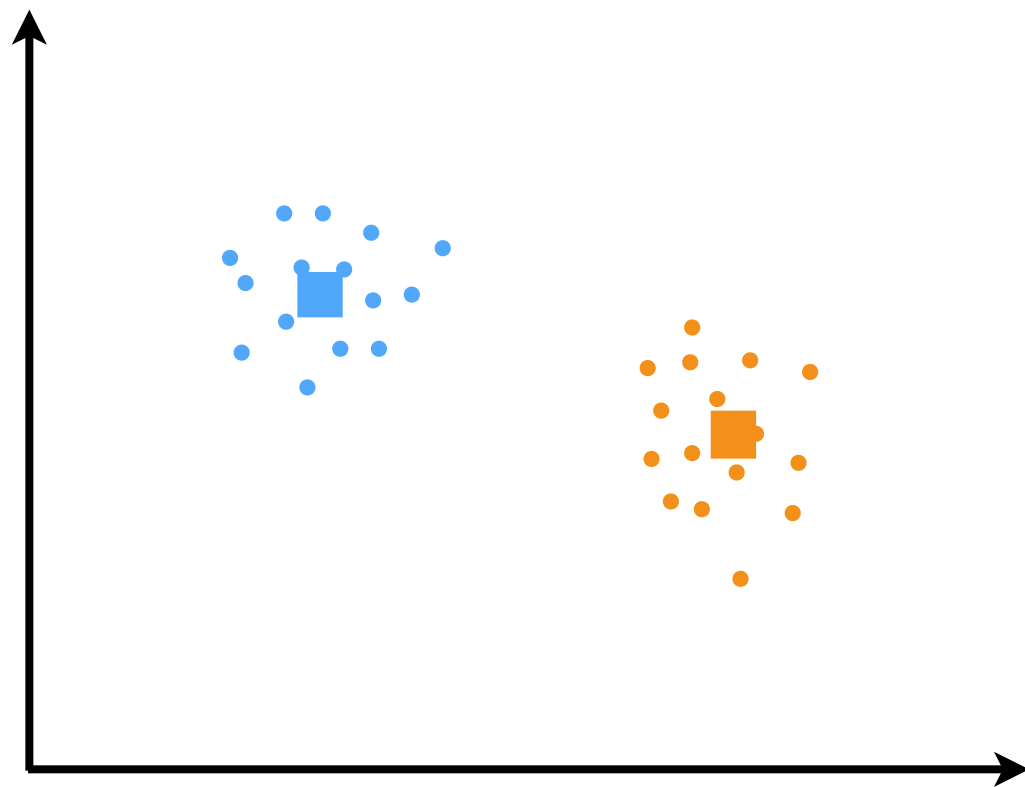


$$\mu_k \stackrel{iid}{\sim} \mathcal{N}(\mu_0, \Sigma_0)$$

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- Finite Gaussian mixture model ($K=2$ clusters)

$$\mu_k \stackrel{iid}{\sim} \mathcal{N}(\mu_0, \Sigma_0)$$



$$\rho_1 \sim \text{Beta}(a_1, a_2)$$

$$\rho_2 = 1 - \rho_1$$

$$z_n$$

$$x_n \stackrel{indep}{\sim} \mathcal{N}(\mu_{z_n}, \Sigma)$$

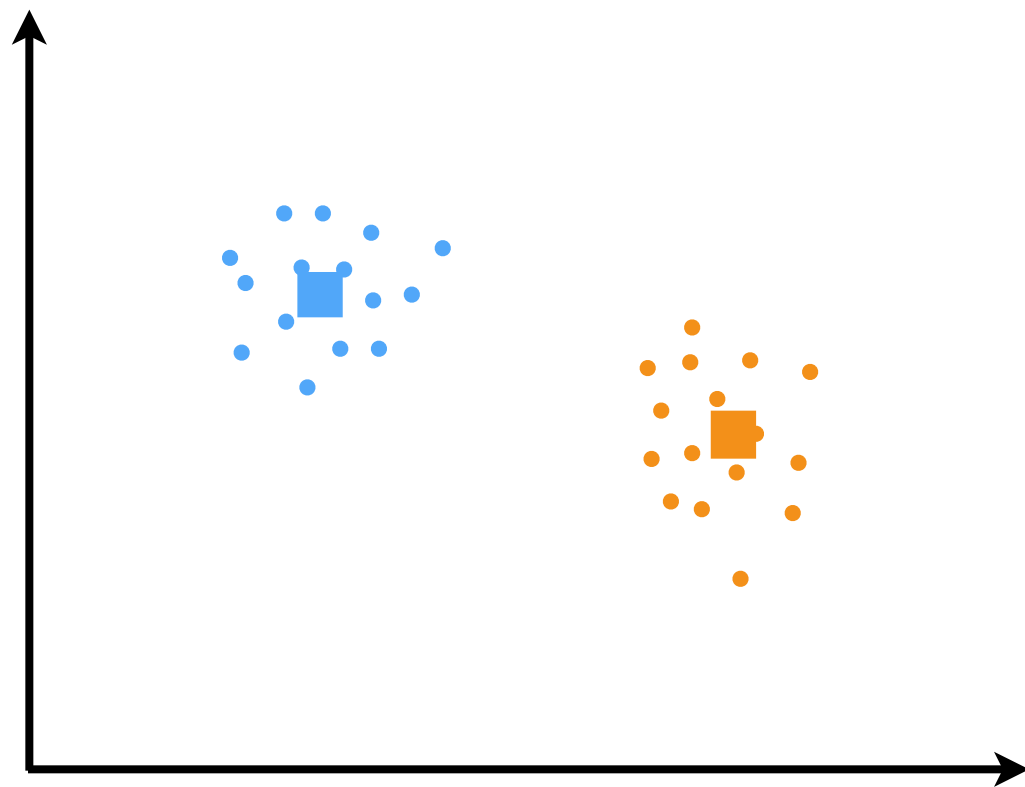


ρ_1

ρ_2

Generative model

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$$\rho_2 = 1 - \rho_1$$

□ $z_n \stackrel{iid}{\sim} \text{Categorical}(\rho_1, \rho_2)$

$$x_n \stackrel{indep}{\sim} \mathcal{N}(\mu_{z_n}, \Sigma)$$

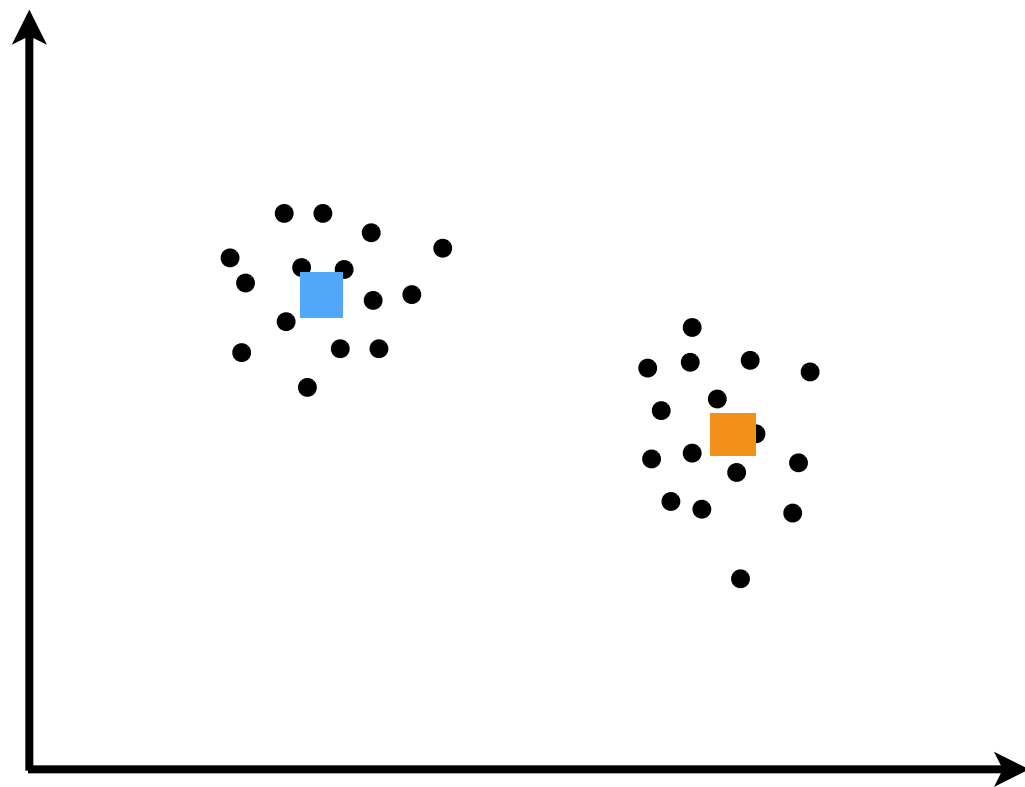


ρ_1

ρ_2

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ρ_1

ρ_2



Beta distribution review

$$\text{Beta}(\rho_1 | a_1, a_2) = \frac{\Gamma(a_1 + a_2)}{\Gamma(a_1)\Gamma(a_2)} \rho_1^{a_1-1} (1 - \rho_1)^{a_2-1}$$

$$\rho_1 \in (0, 1)$$

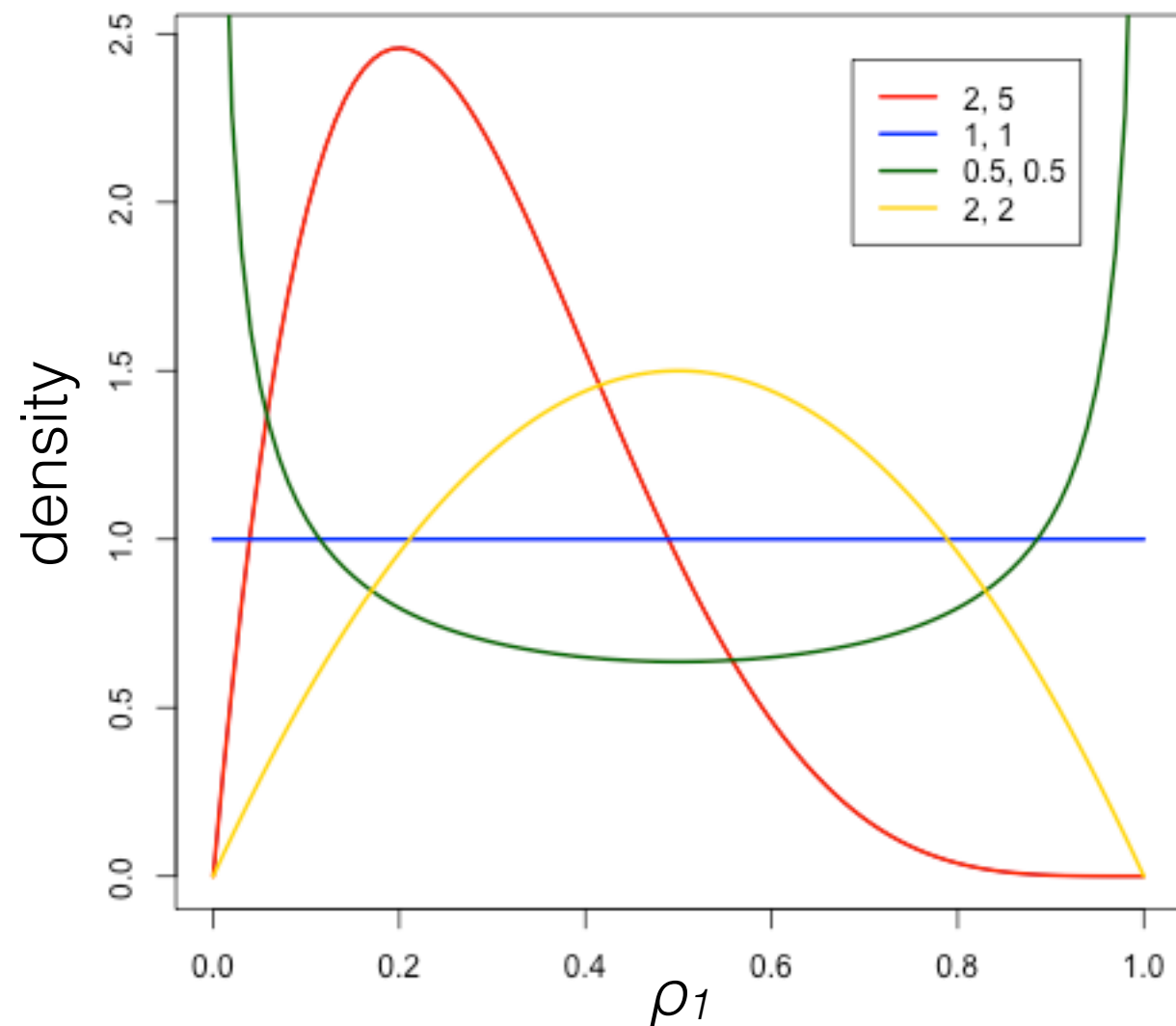
$$a_1, a_2 > 0$$

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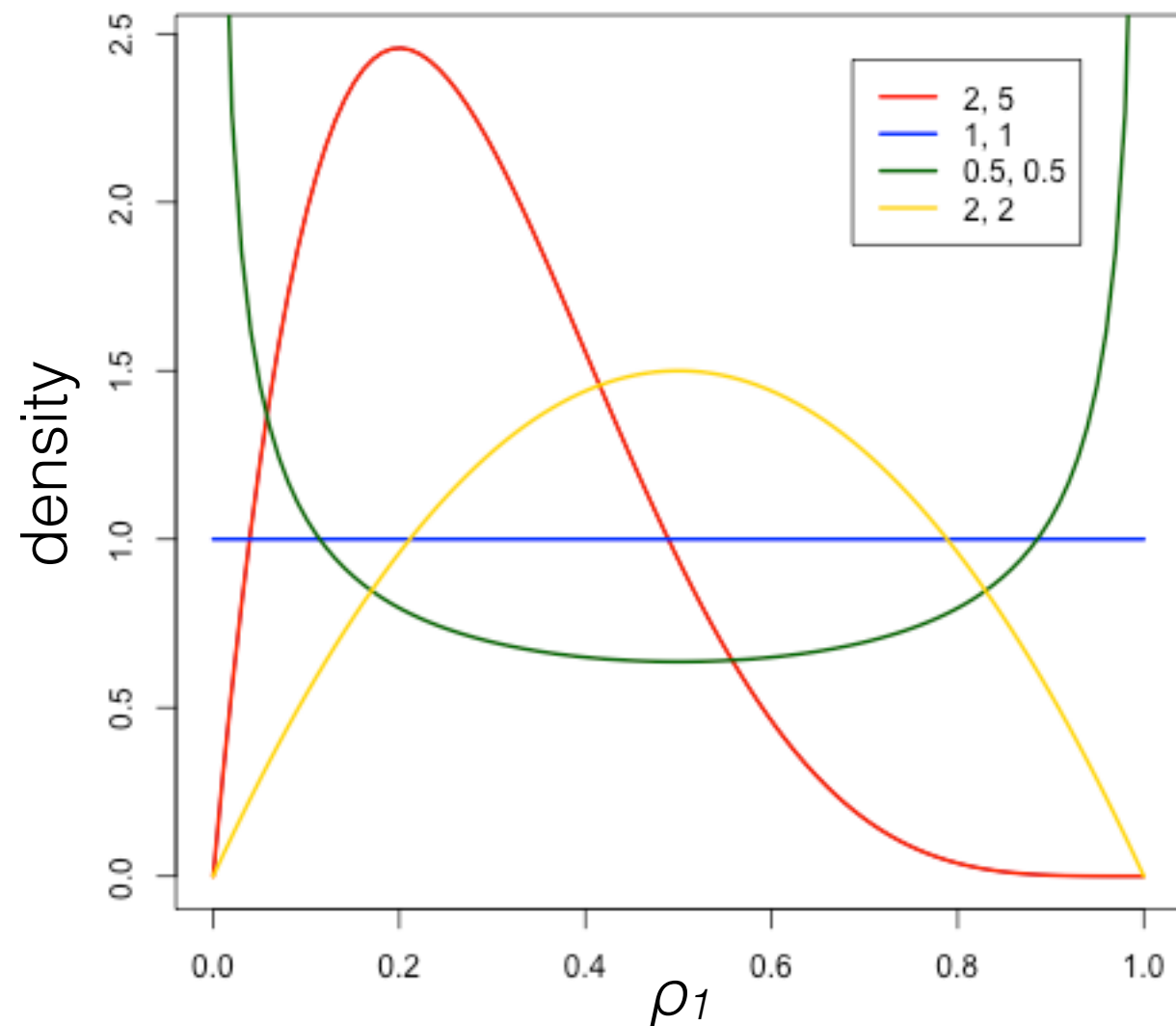
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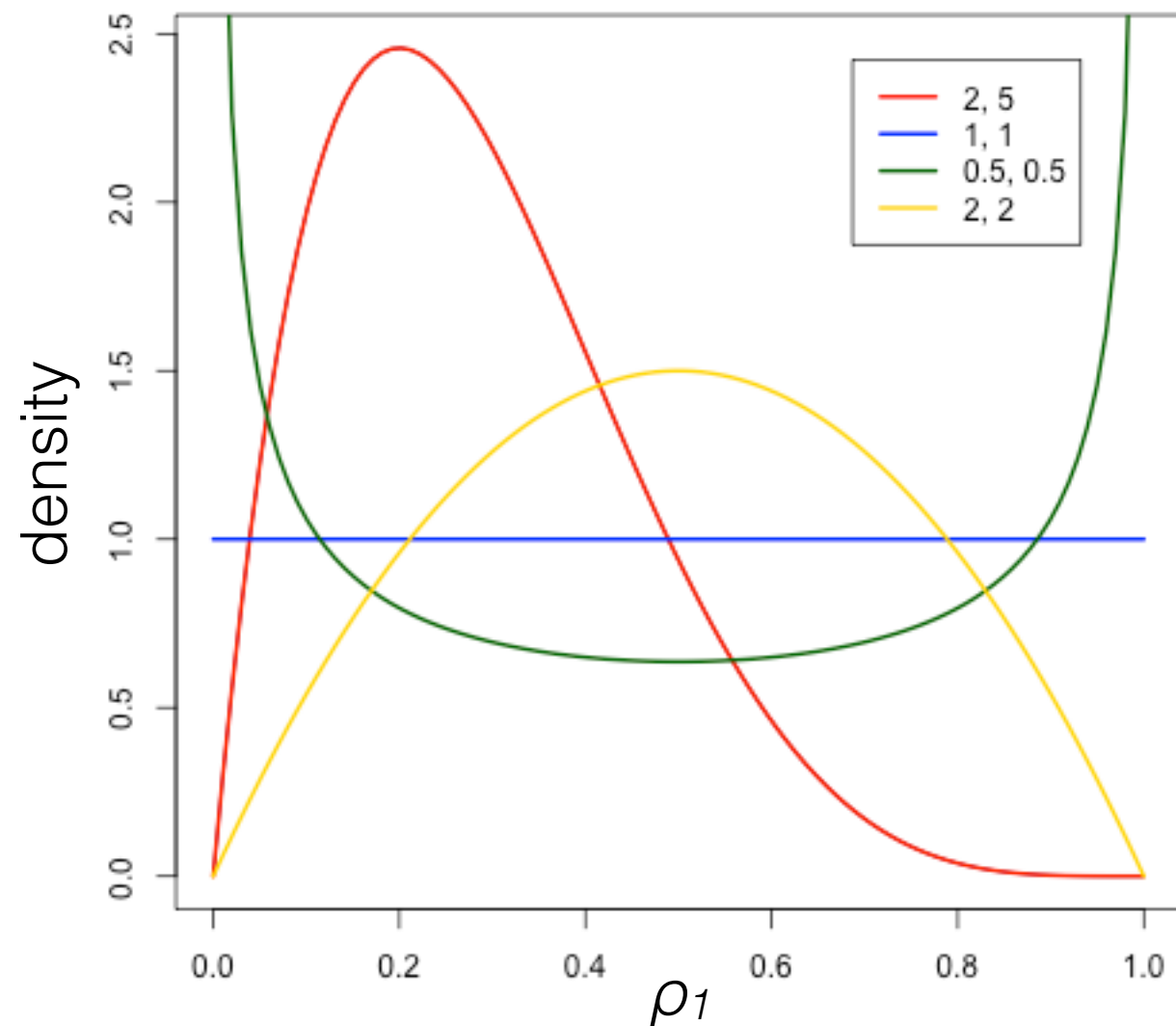


- What happens?



Beta distribution review

$$\text{Beta}(\rho_1 | a_1, a_2) = \frac{\Gamma(a_1 + a_2)}{\Gamma(a_1)\Gamma(a_2)} \rho_1^{a_1-1} (1 - \rho_1)^{a_2-1} \quad \begin{array}{l} \rho_1 \in (0, 1) \\ a_1, a_2 > 0 \end{array}$$



- What happens?

$$a = a_1 = a_2 \rightarrow 0$$

$$a = a_1 = a_2 \rightarrow \infty$$

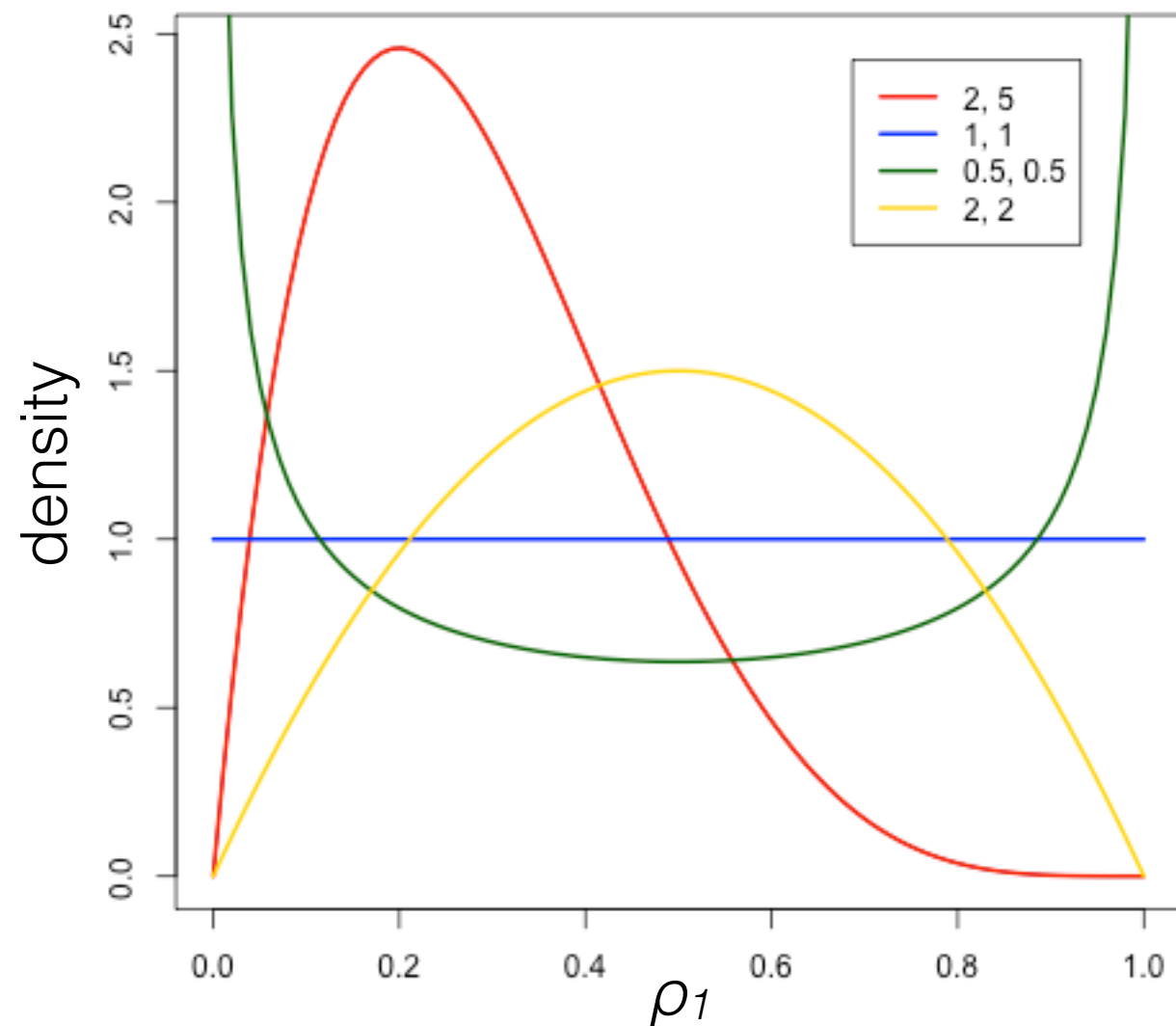
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Beta distribution review

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- What happens?

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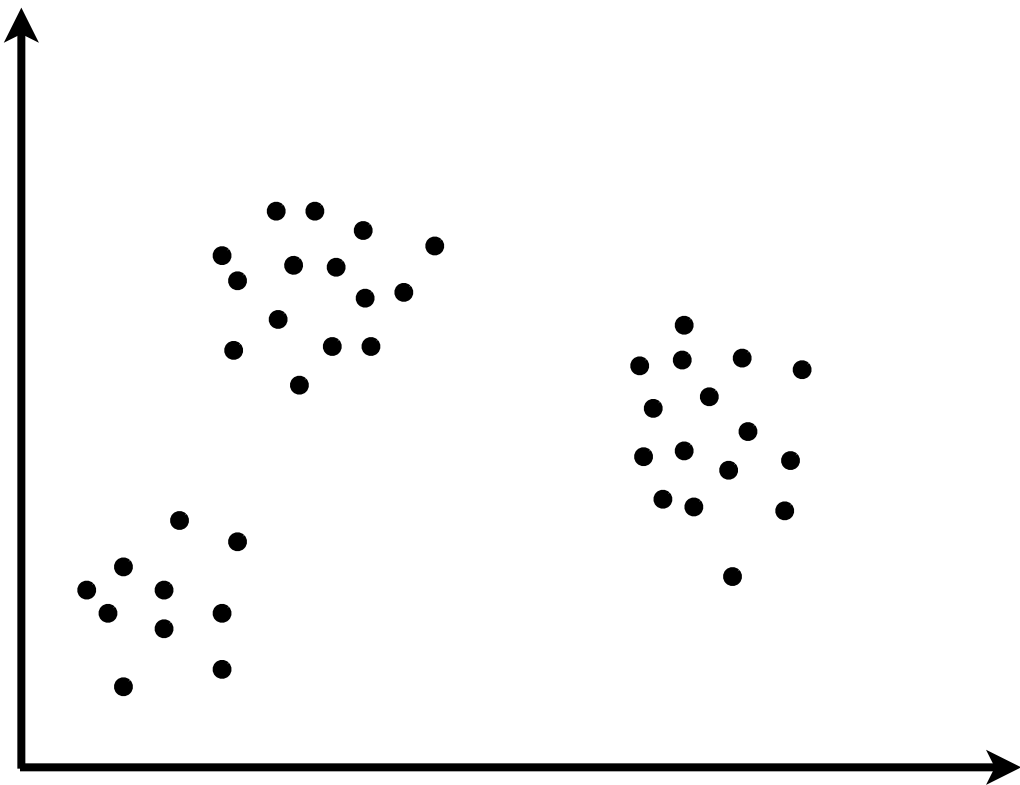
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[demo]

Generative model

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- Finite Gaussian mixture model (K clusters)



Generative model

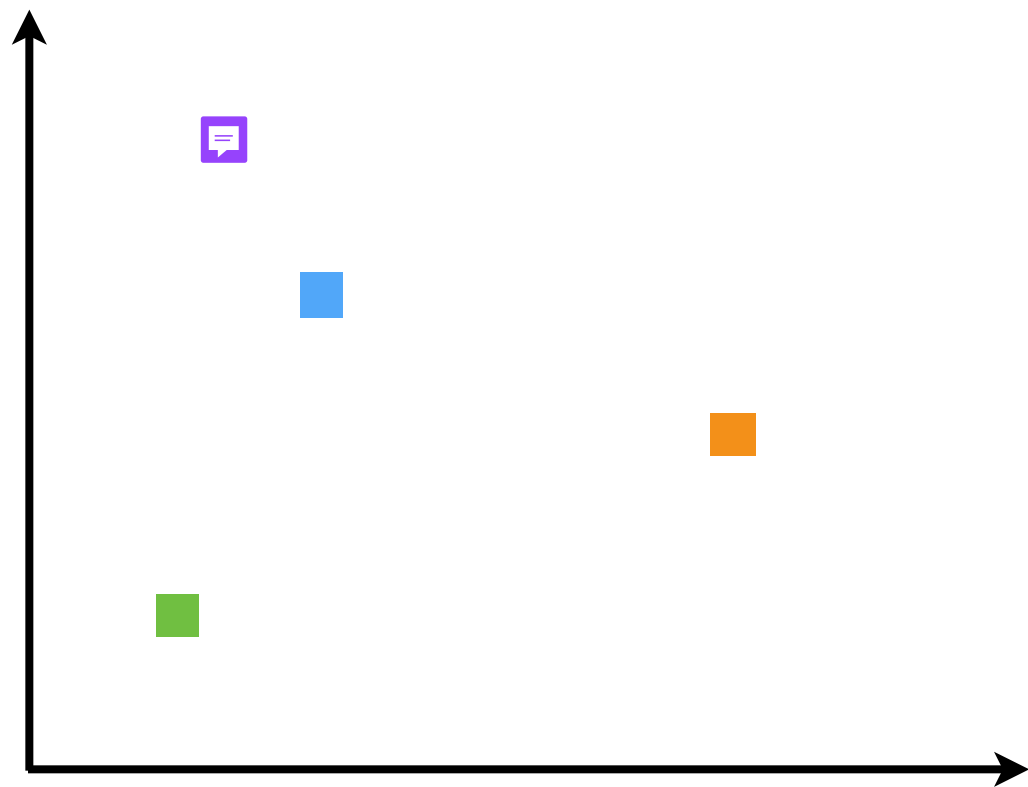
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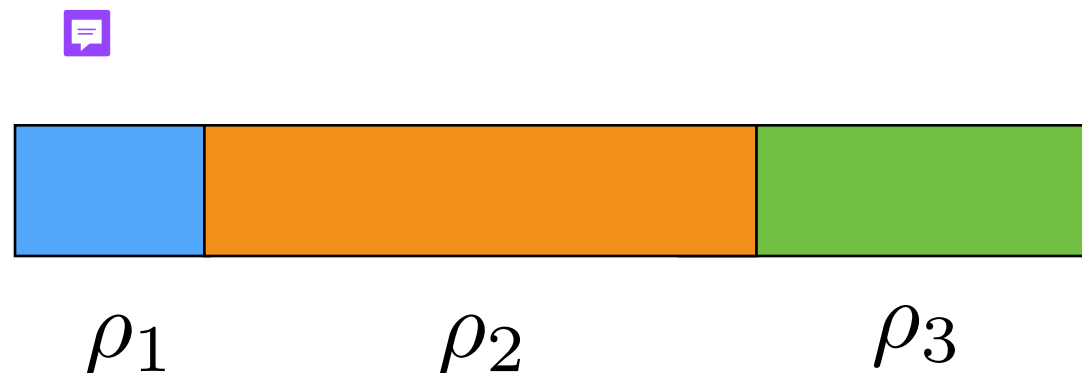
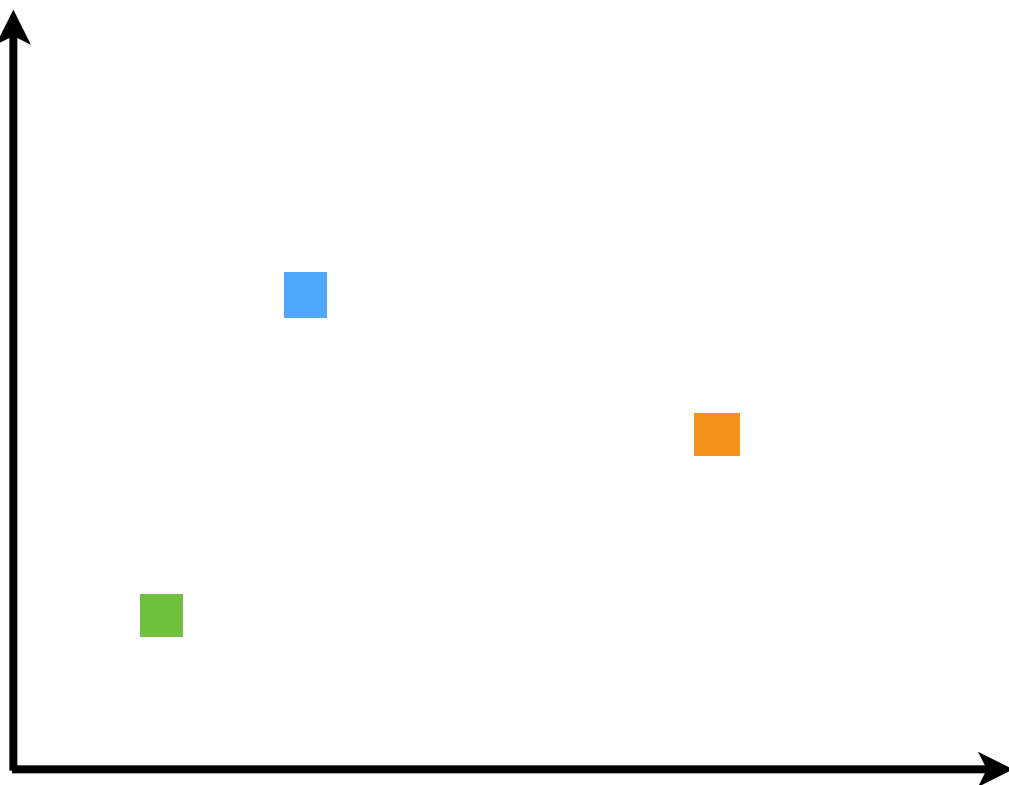
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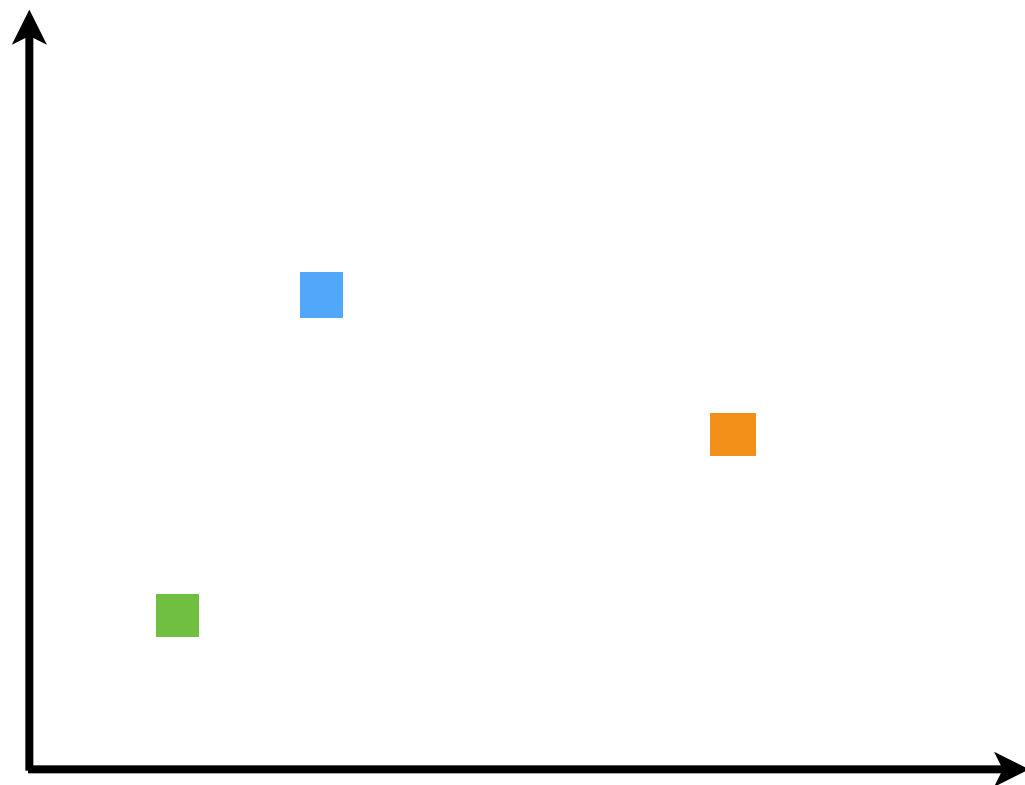
$$\mu_k \stackrel{iid}{\sim} \mathcal{N}(\mu_0, \Sigma_0)$$

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K})$$



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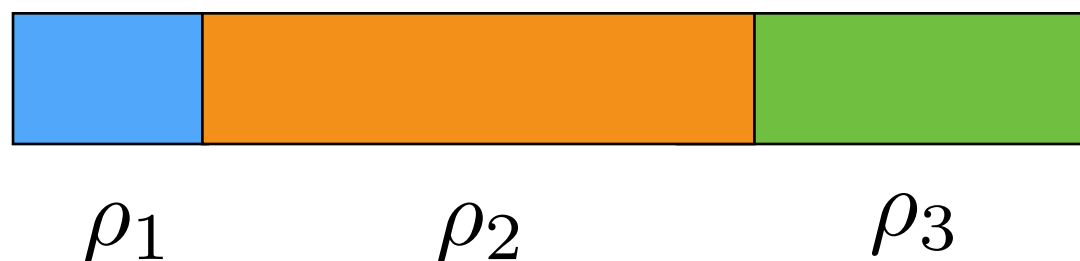
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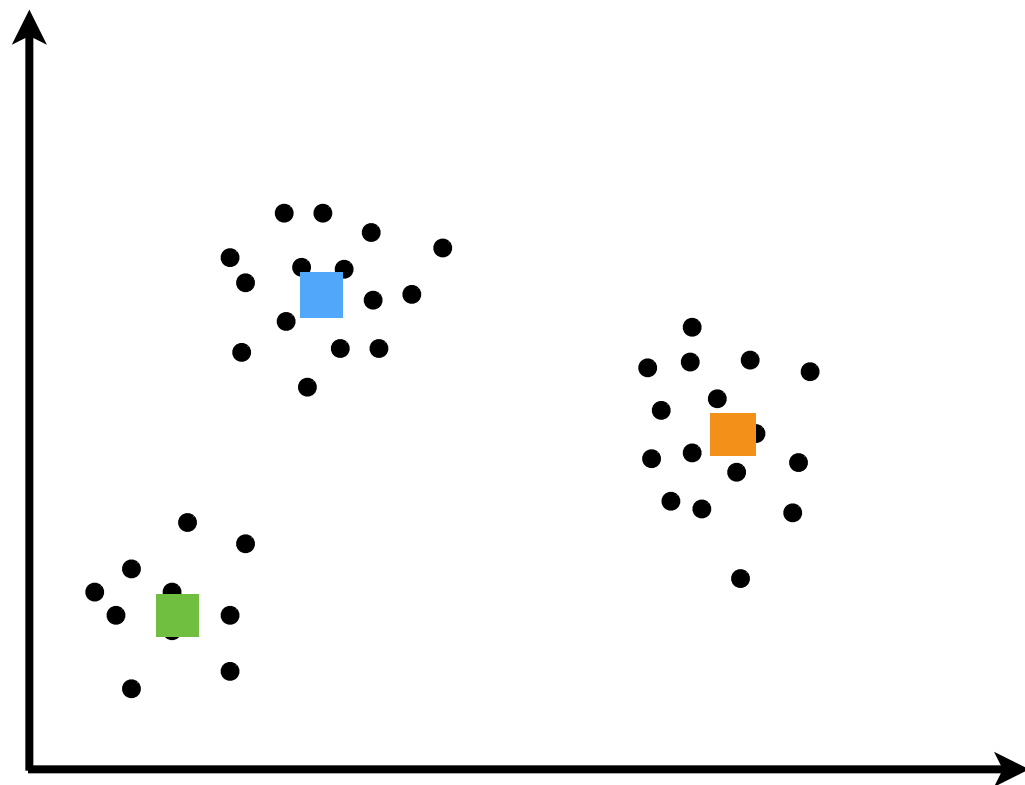


$$z_n \stackrel{iid}{\sim} \text{Categorical}(\rho_{1:K})$$



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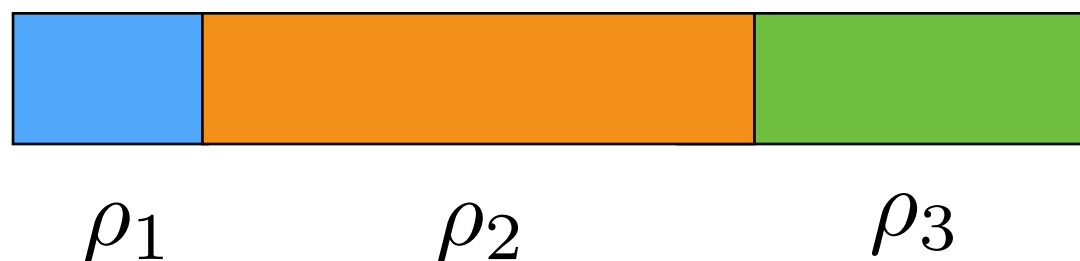
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$$z_n \stackrel{iid}{\sim} \text{Categorical}(\rho_{1:K})$$

$$x_n \stackrel{indep}{\sim} \mathcal{N}(\mu_{z_n}, \Sigma)$$



Dirichlet distribution review

$$\text{Dirichlet}(\rho_{1:K} | a_{1:K}) = \frac{\Gamma(\sum_{k=1}^K a_k)}{\prod_{k=1}^K \Gamma(a_k)} \prod_{k=1}^K \rho_k^{a_k - 1} \quad a_k > 0$$



Dirichlet distribution review

$$\text{Dirichlet}(\rho_{1:K} | a_{1:K}) = \frac{\Gamma(\sum_{k=1}^K a_k)}{\prod_{k=1}^K \Gamma(a_k)} \prod_{k=1}^K \rho_k^{a_k - 1}$$

$$a_k > 0$$

$$\rho_k \in (0, 1)$$

$$\sum_k \rho_k = 1$$

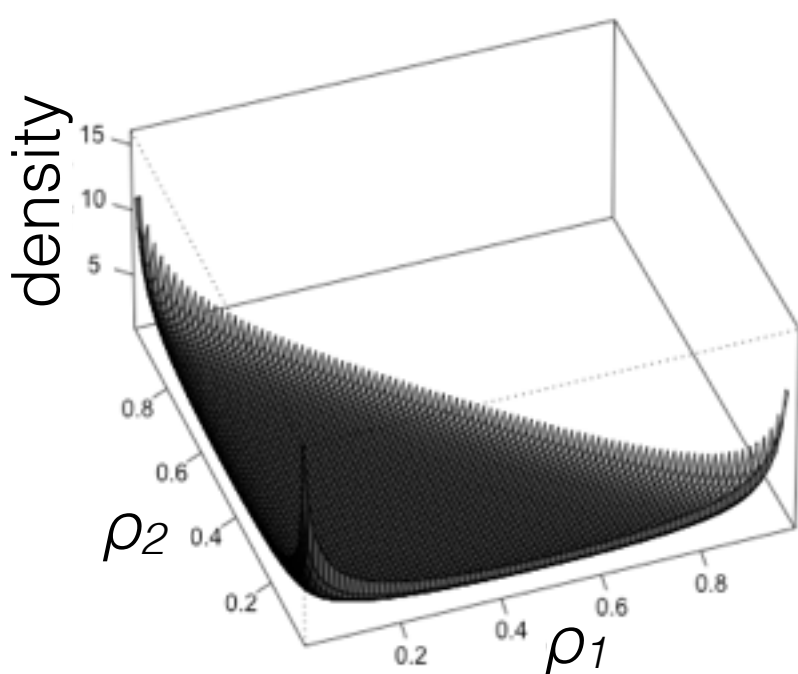


Dirichlet distribution review

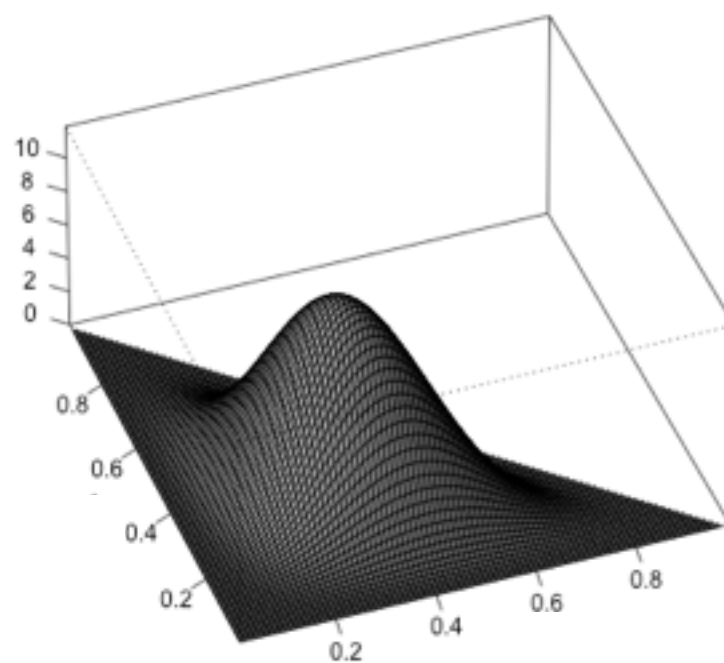
$$\text{Dirichlet}(\rho_{1:K} | a_{1:K}) = \frac{\Gamma(\sum_{k=1}^K a_k)}{\prod_{k=1}^K \Gamma(a_k)} \prod_{k=1}^K \rho_k^{a_k - 1} \quad \begin{array}{l} a_k > 0 \\ \rho_k \in (0, 1) \\ \sum_k \rho_k = 1 \end{array}$$



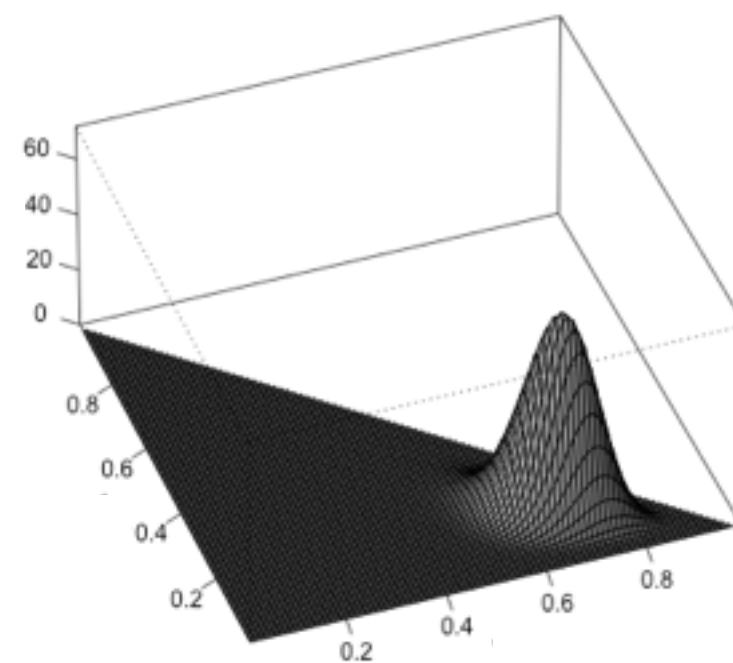
$a = (0.5, 0.5, 0.5)$



$a = (5, 5, 5)$



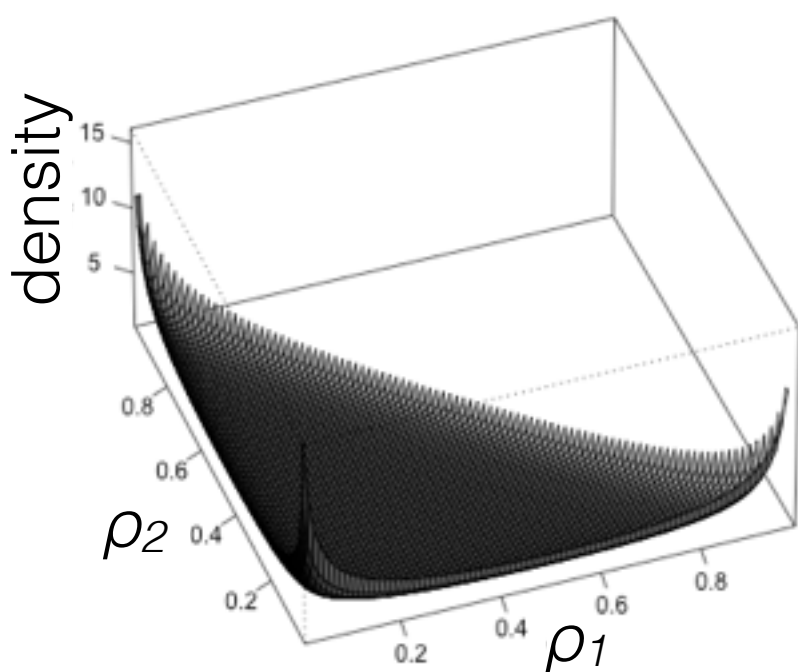
$a = (40, 10, 10)$



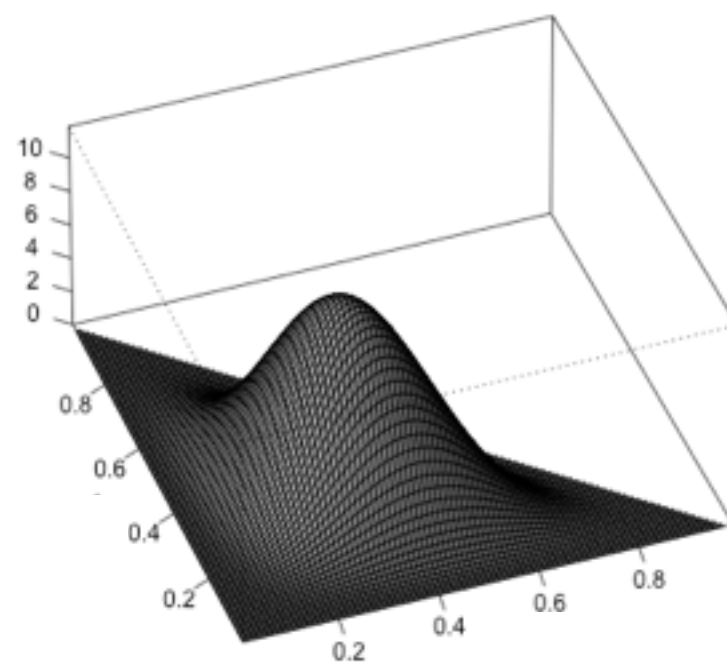
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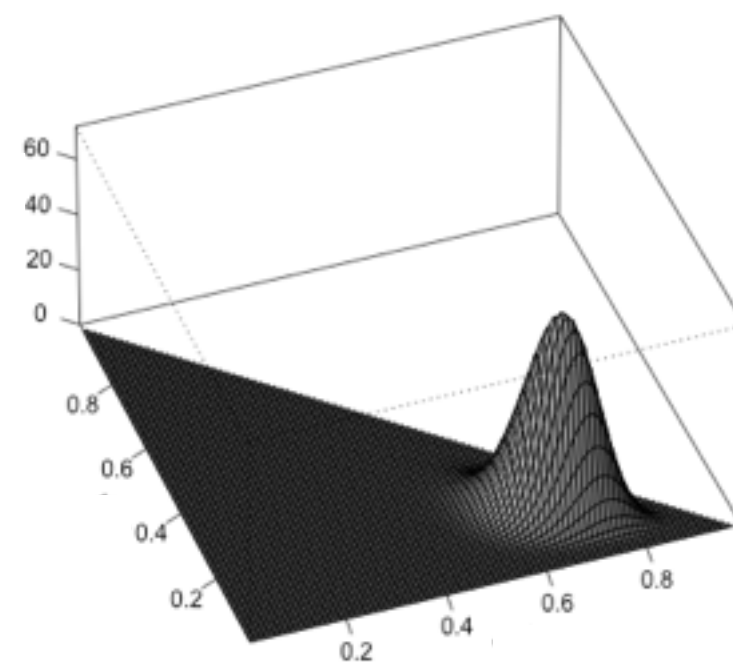
$a = (0.5, 0.5, 0.5)$



$a = (5, 5, 5)$



$a = (40, 10, 10)$



- What happens?

$$a = a_k = 1$$

$$a = a_k \rightarrow 0$$

$$a = a_k \rightarrow \infty$$



[demo]

So far $K \ll N$. What if not?



So far $K \ll N$. What if not?



ρ_1

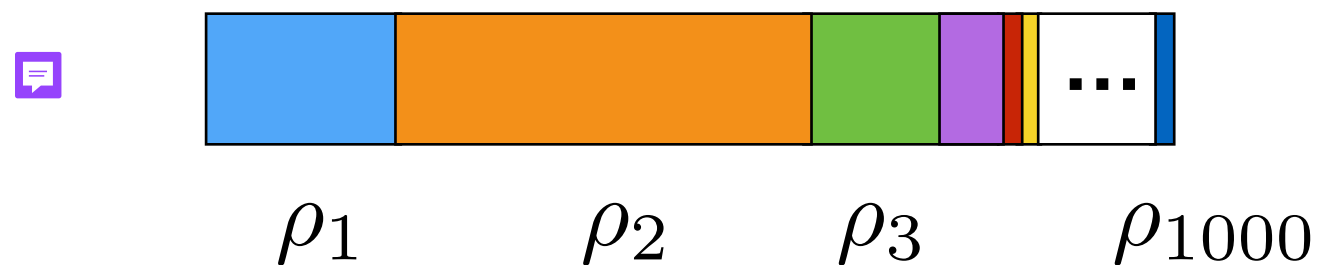
ρ_2

ρ_3

ρ_{1000}

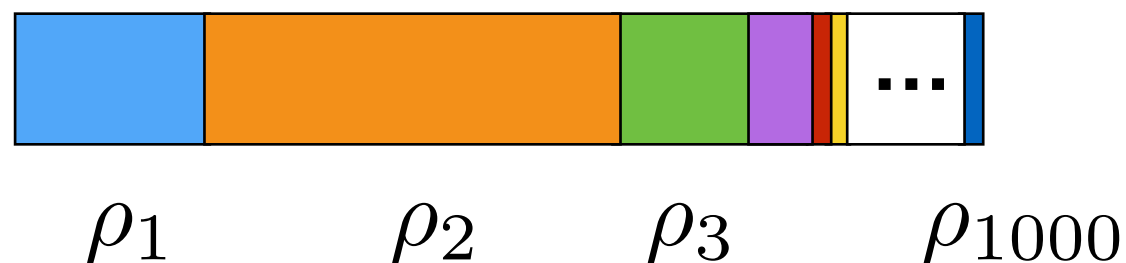
So far $K \ll N$. What if not?

- e.g. species sampling, topic modeling, groups on a social network, etc.



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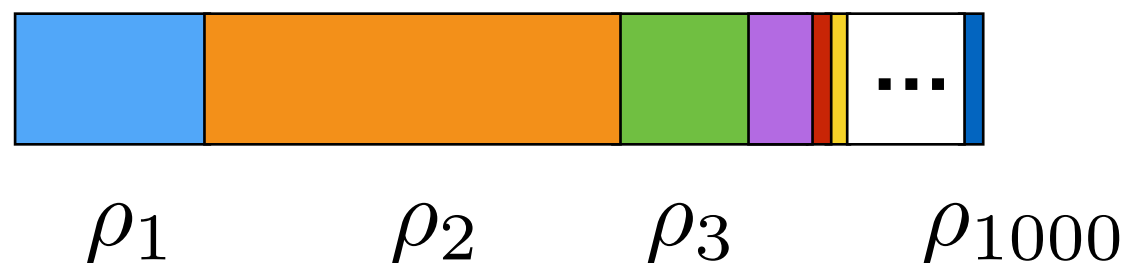
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- Components: number of latent groups

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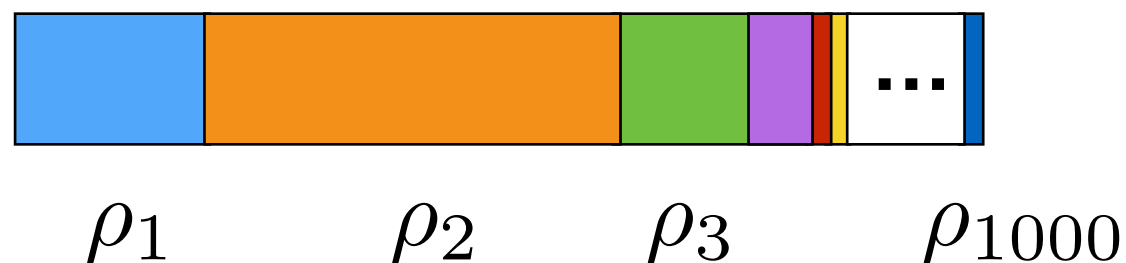
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- Components: number of latent groups
- Clusters: number of components represented in the data

So far $K \ll N$. What if not?

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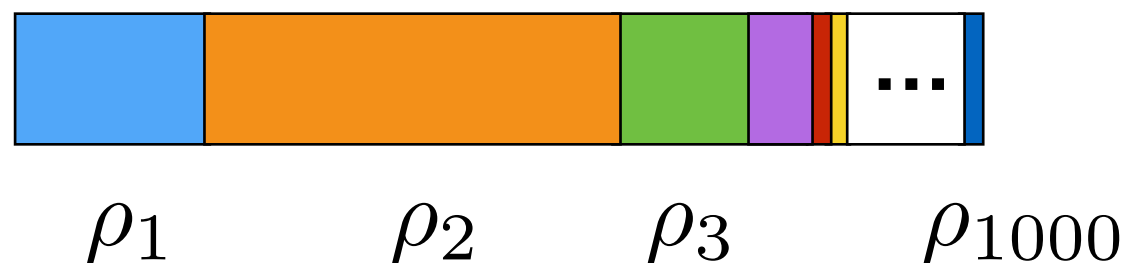


- Components: number of latent groups
- Clusters: number of components represented in the data
- [demo 1, demo 2]



So far $K \ll N$. What if not?

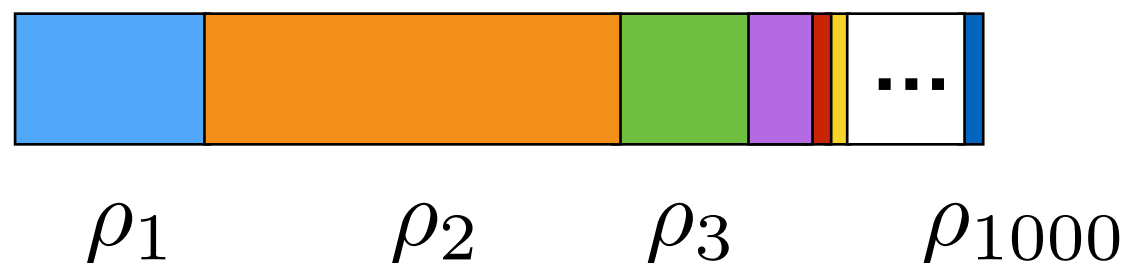
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- Components: number of latent groups
- Clusters: number of components represented in the data
- [demo 1, demo 2]
- Number of clusters is random

So far $K \ll N$. What if not?

- e.g. species sampling, topic modeling, groups on a social network, etc.



- Components: number of latent groups
- Clusters: number of components represented in the data
- [demo 1, demo 2]
- Number of clusters is random
- Number of clusters grows with N

- Here, difficult to choose finite K in advance (contrast with small K): don't know K , difficult to infer, streaming data



Choosing $K = \infty$

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- “Stick breaking”

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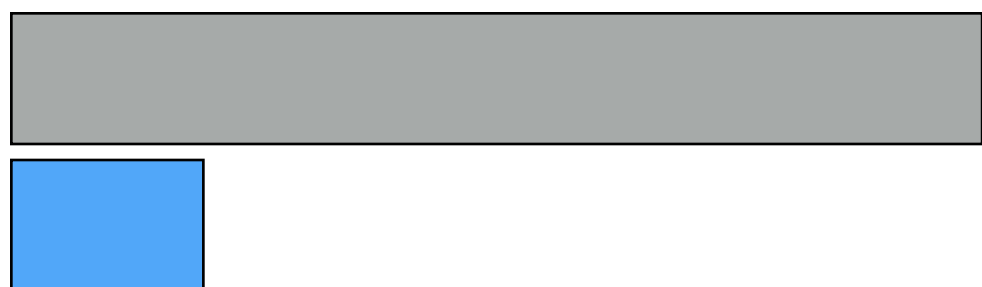
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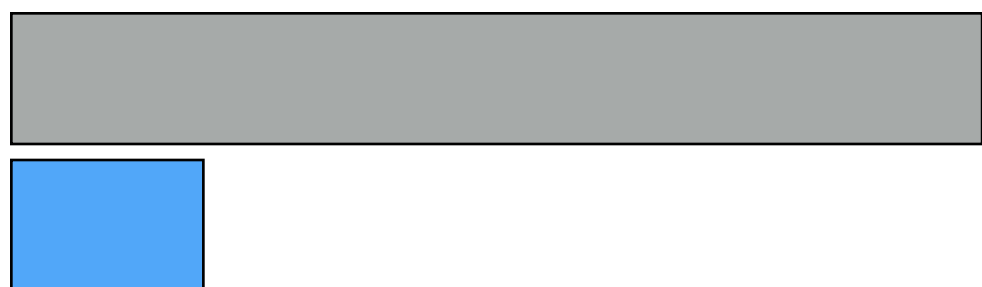
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$$\rho_1 = V_1$$

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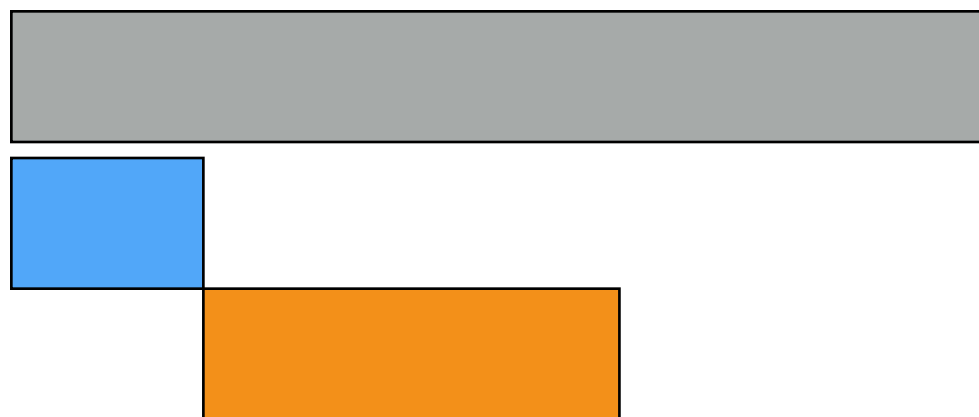
$$\rho_1 = V_1$$

$$V_2 \sim \text{Beta}(a_2, a_3 + a_4)$$

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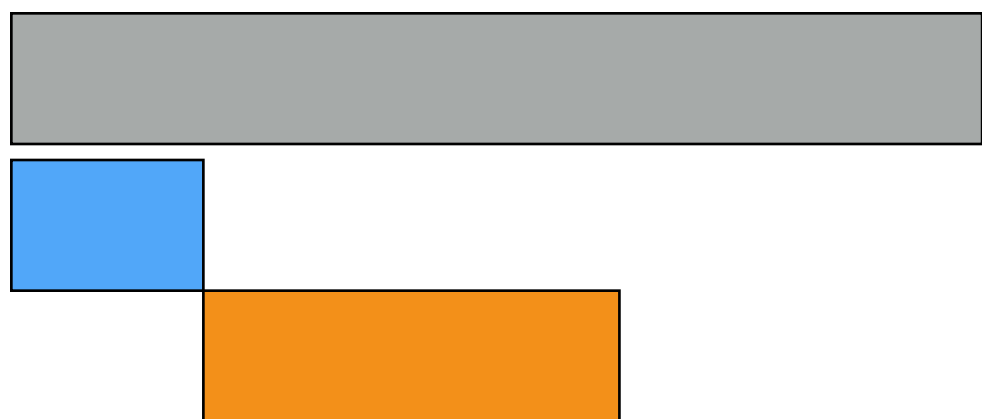
$$V_2 \sim \text{Beta}(a_2, a_3 + a_4)$$

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$$V_2 \sim \text{Beta}(a_2, a_3 + a_4)$$

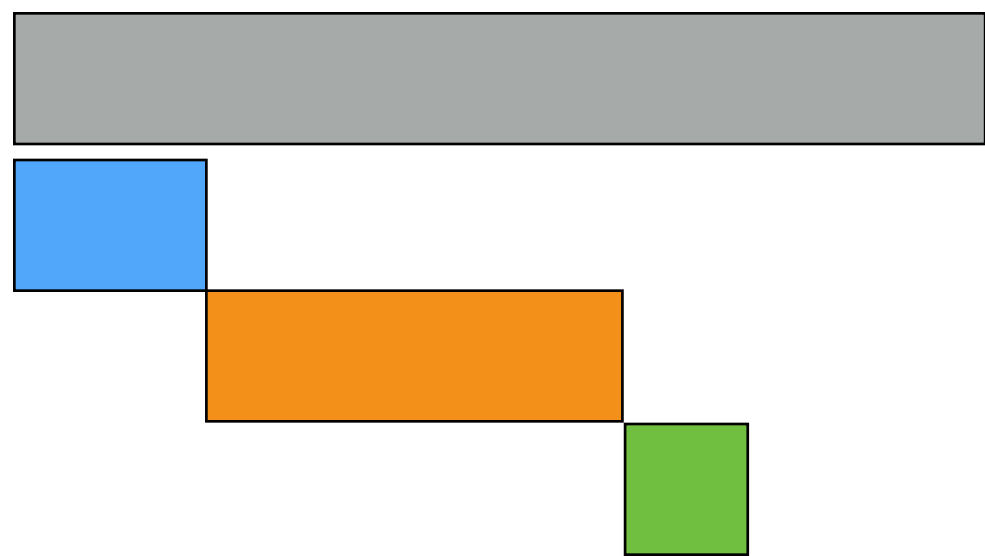
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- “Stick breaking”

$$V_1 \sim \text{Beta}(a_1, a_2 + a_3 + a_4) \quad \rho_1 = V_1$$

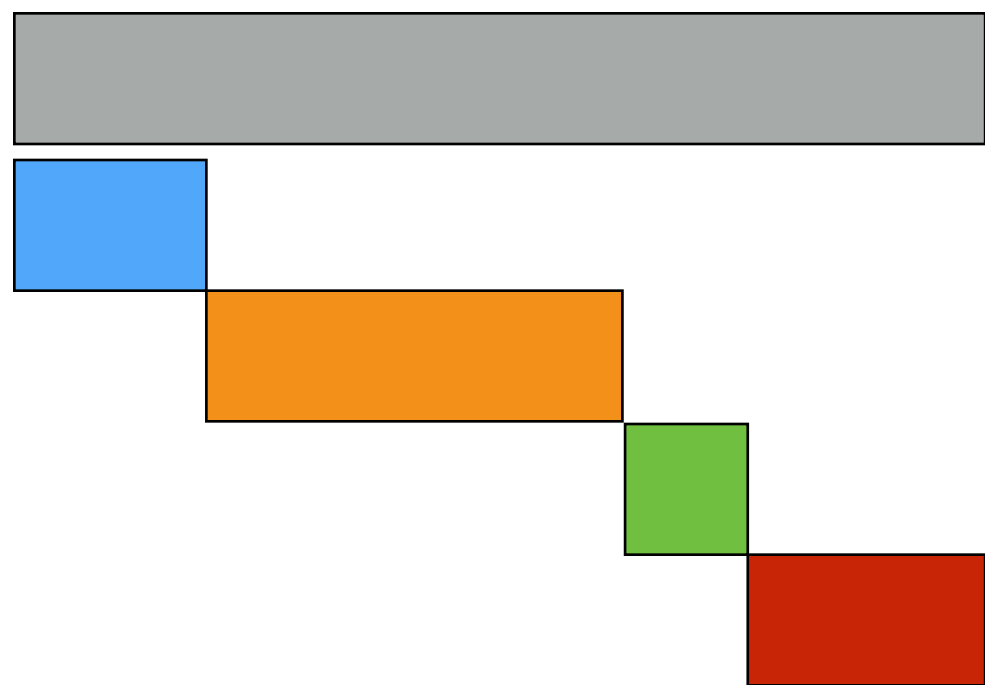
$$V_2 \sim \text{Beta}(a_2, a_3 + a_4) \quad \rho_2 = (1 - V_1)V_2$$

$$V_3 \sim \text{Beta}(a_3, a_4) \quad \rho_3 = (1 - V_1)(1 - V_2)V_3$$

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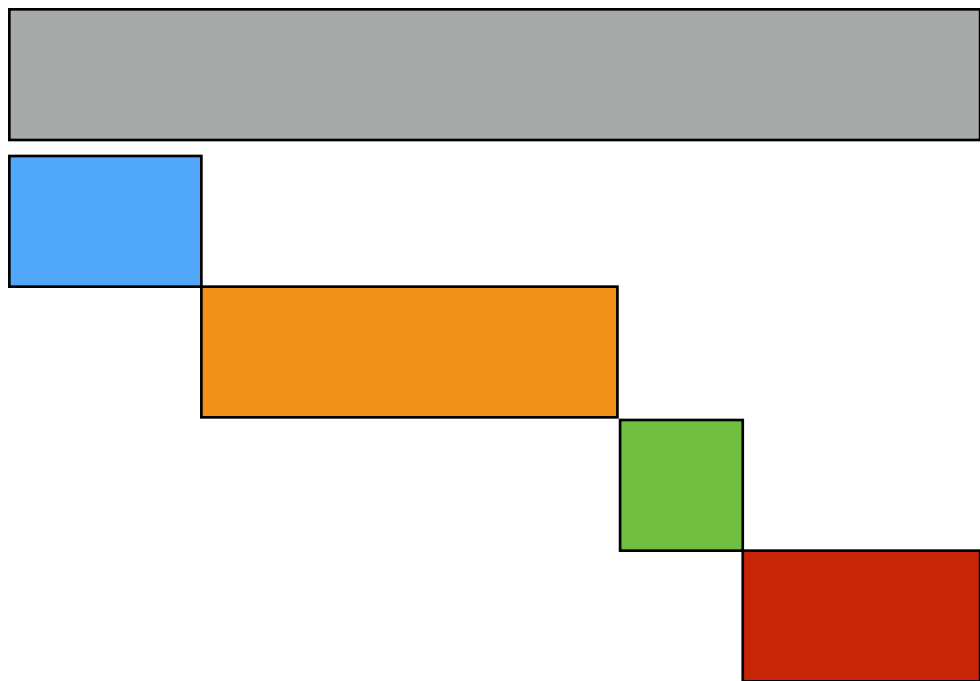
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$$\rho_4 = 1 - \sum_{k=1}^3 \rho_k$$

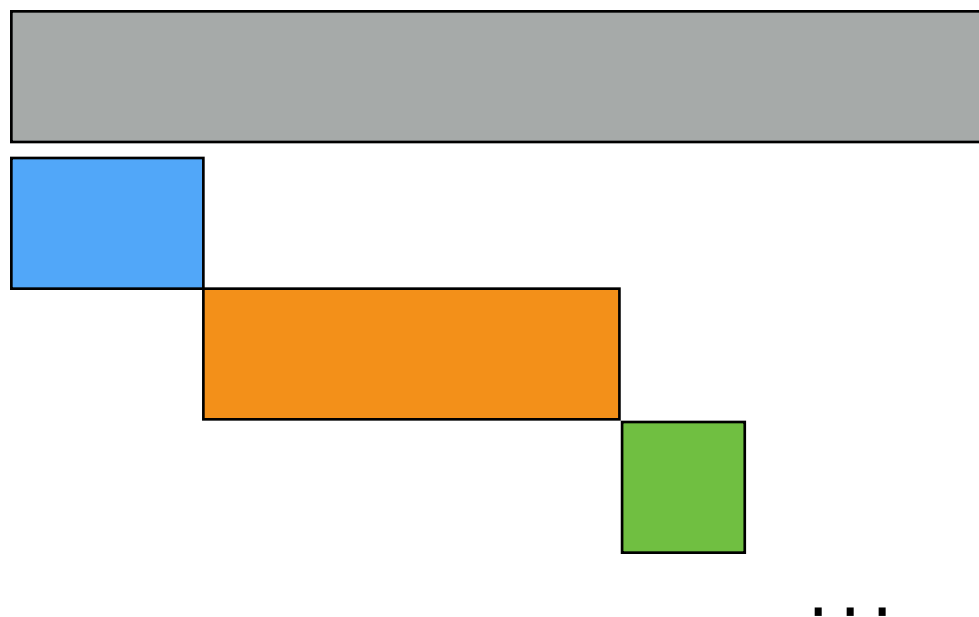
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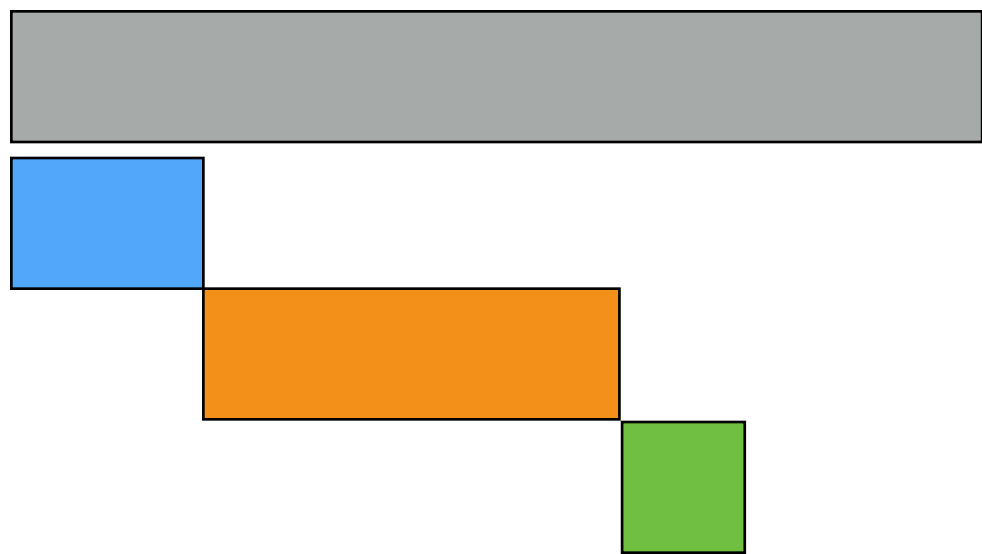
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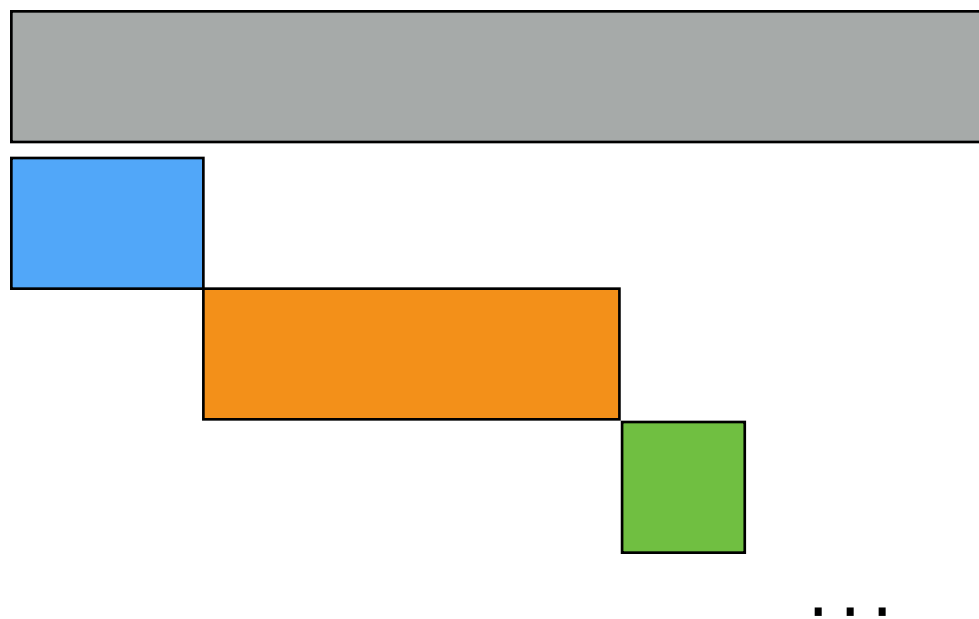
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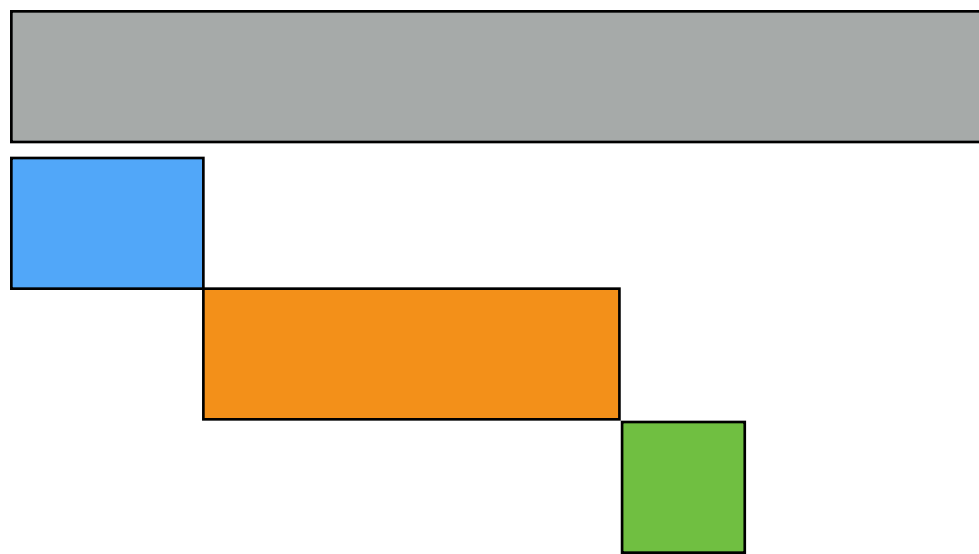
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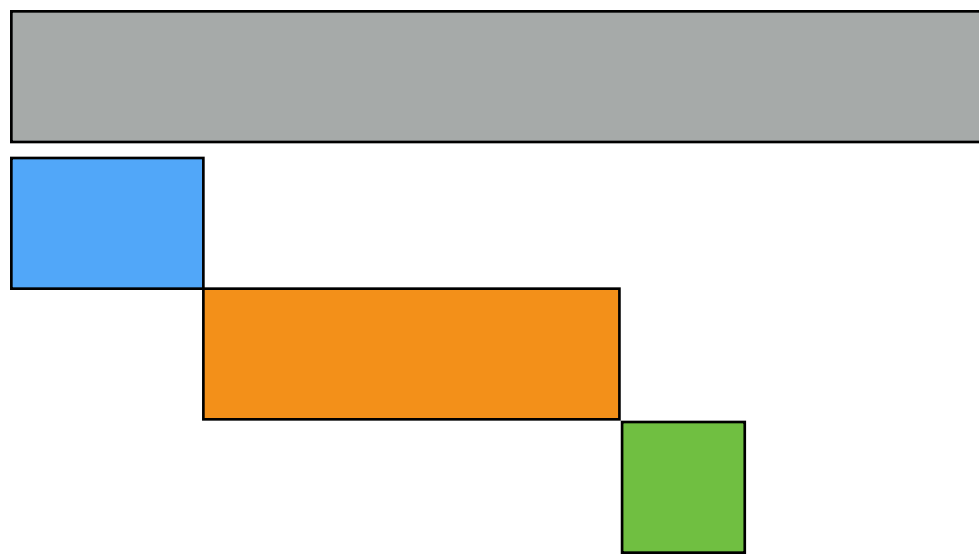
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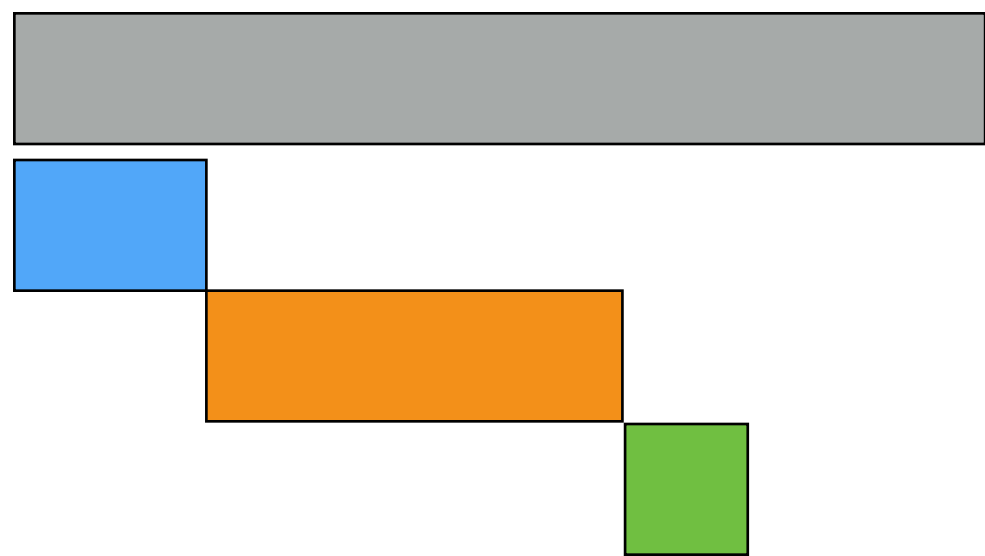
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$$V_k \sim \text{Beta}(a_k, b_k)$$

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$$\rho_1 = V_1$$

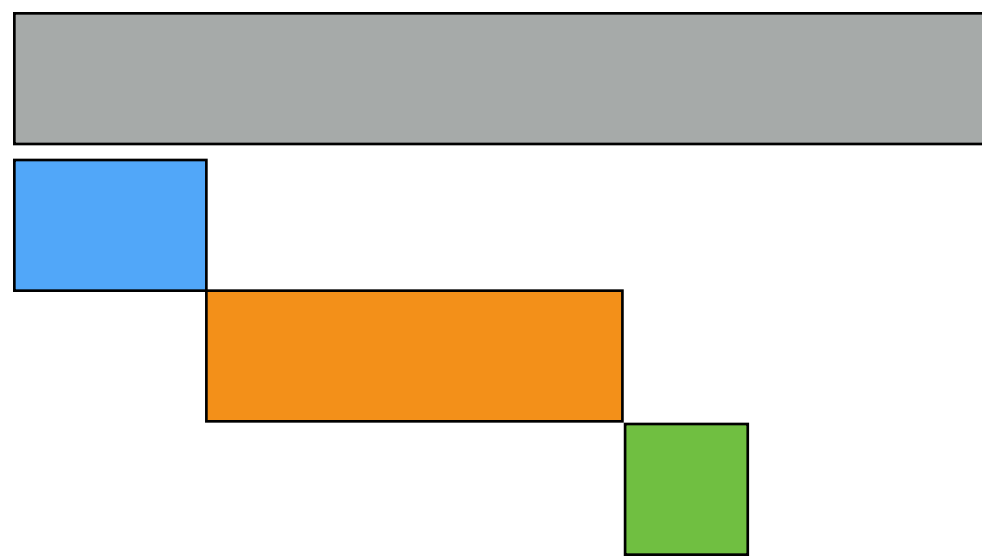
$$\rho_2 = (1 - V_1)V_2$$

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[Ishwaran, James 2001]

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...

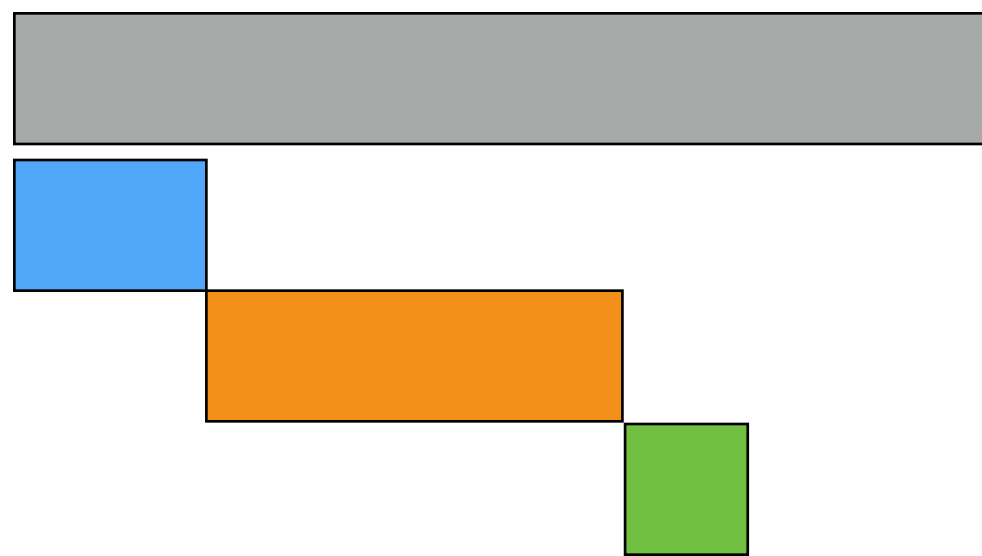
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- **Dirichlet process stick-breaking**: $a_k = 1, b_k = \alpha > 0$
- Griffiths-Engen-McCloskey (**GEM**) distribution: 
 $\rho = (\rho_1, \rho_2, \dots) \sim \text{GEM}(\alpha)$



$$V_1 \sim \text{Beta}(a_1, b_1)$$

$$\rho_1 = V_1$$

$$V_2 \sim \text{Beta}(a_2, b_2)$$

$$\rho_2 = (1 - V_1)V_2$$

...

$$V_k \sim \text{Beta}(a_k, b_k)$$

$$\rho_k = \left[\prod_{j=1}^{k-1} (1 - V_j) \right] V_k$$

Choosing $K = \infty$

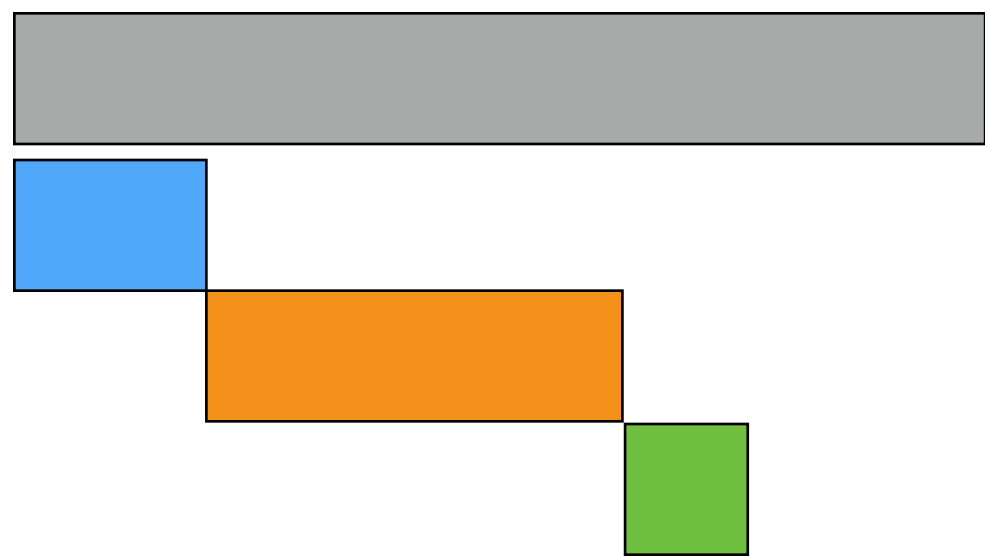
- Here, difficult to choose finite K in advance (contrast with small K): don't know K , difficult to infer, streaming data
- How to generate $K = \infty$ strictly positive frequencies that sum to one?

- **Dirichlet process stick-breaking**: $a_k = 1, b_k = \alpha > 0$

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$$\rho = (\rho_1, \rho_2, \dots) \sim \text{GEM}(\alpha)$$

 [demo]



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Dirichlet process mixture model



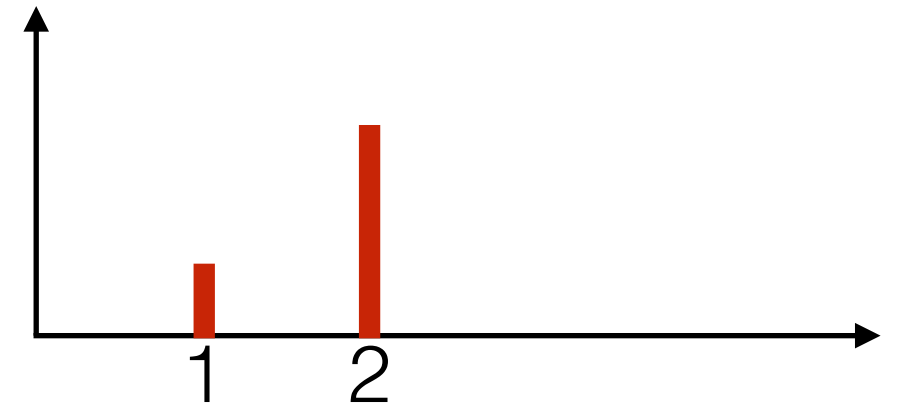
Dirichlet process mixture model

- Gaussian mixture model

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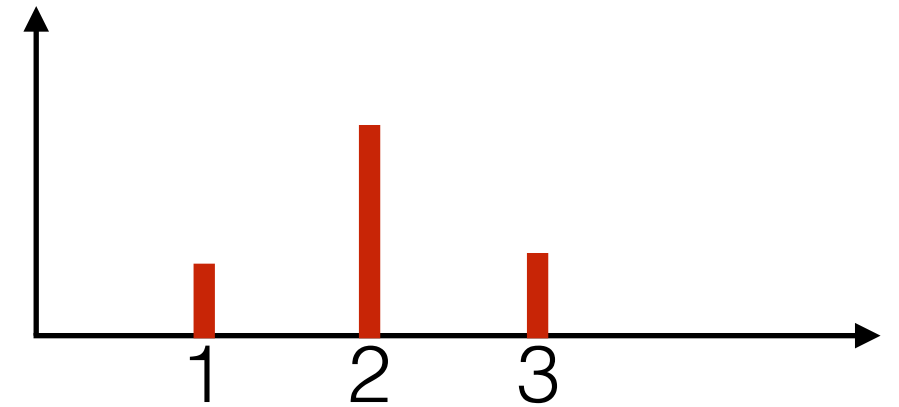
$$\rho = (\rho_1, \rho_2) \sim \text{Beta}(a_1, a_2)$$



Dirichlet process mixture model

- Gaussian mixture model

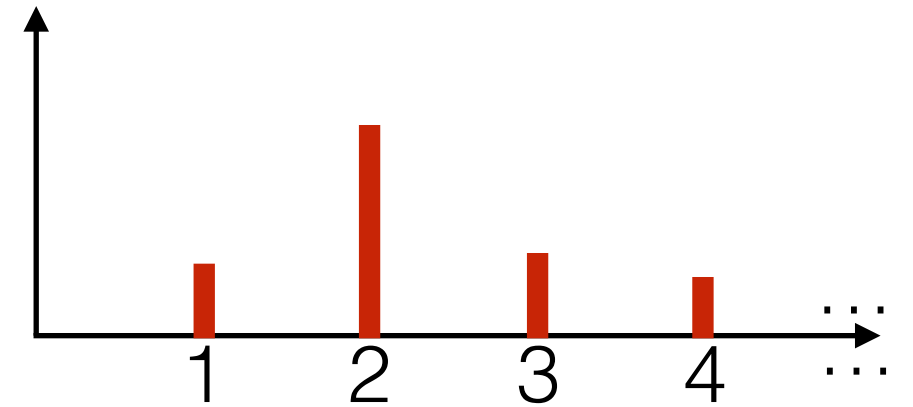
$$\rho = (\rho_1, \dots, \rho_K) \sim \text{Dir}(a_{1:K})$$



Dirichlet process mixture model

- Gaussian mixture model

$$\rho = (\rho_1, \rho_2, \dots) \sim \text{GEM}(\alpha)$$

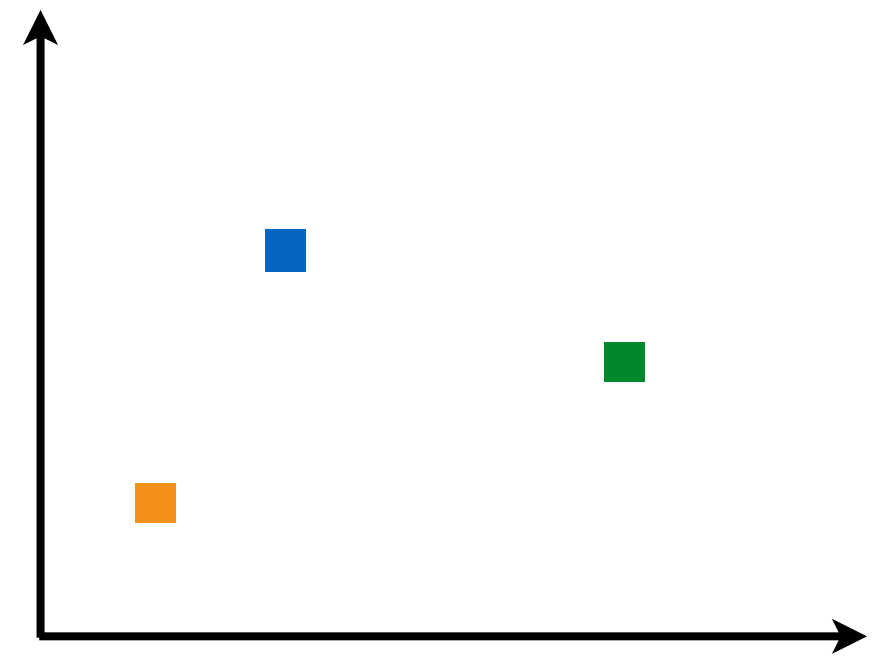
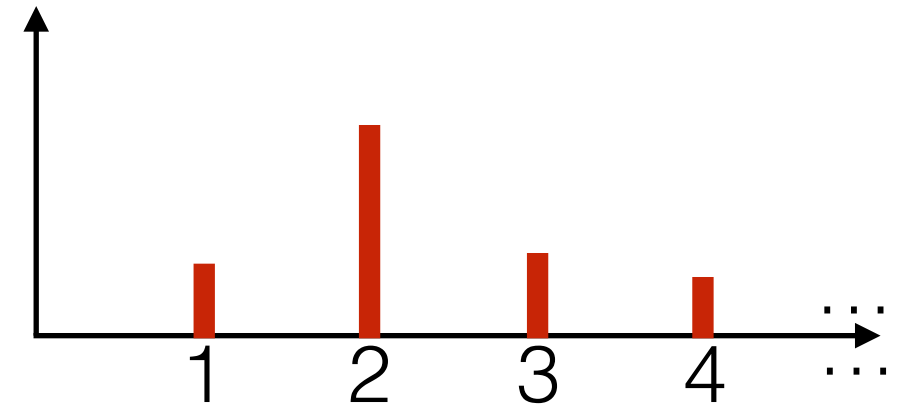


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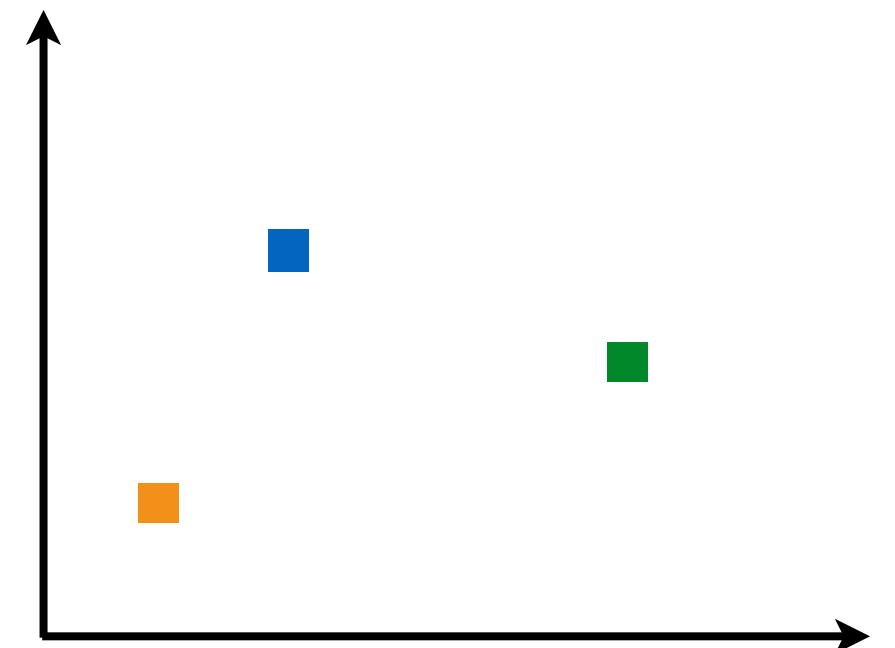
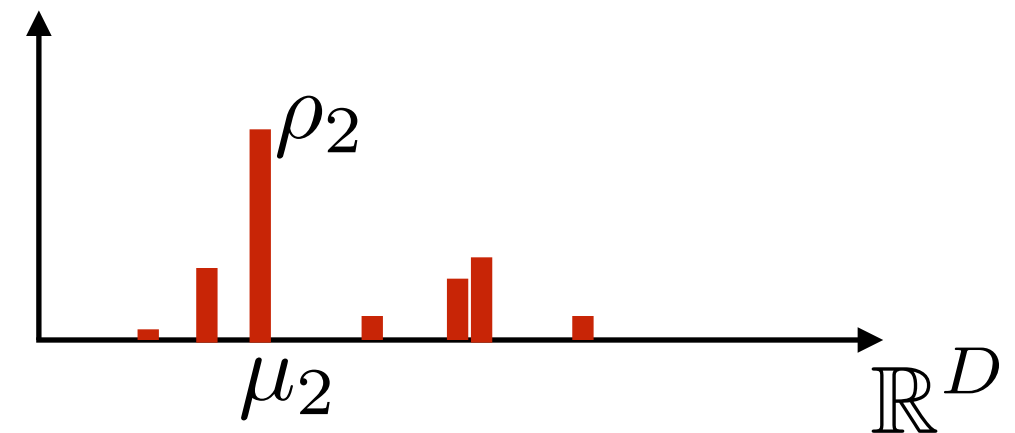
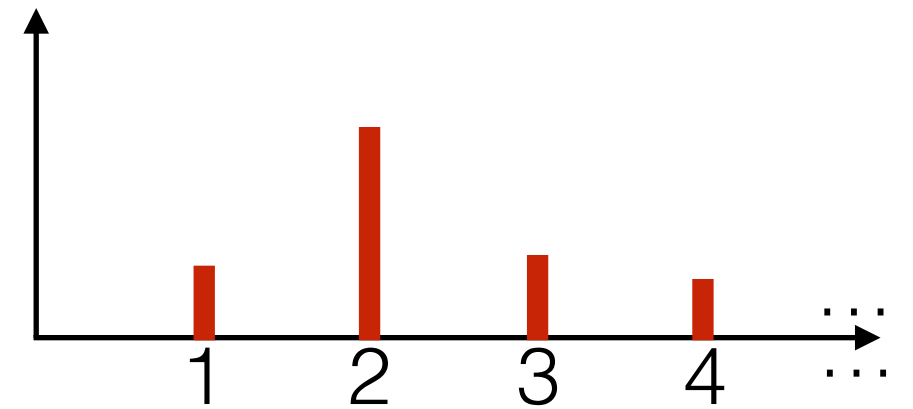


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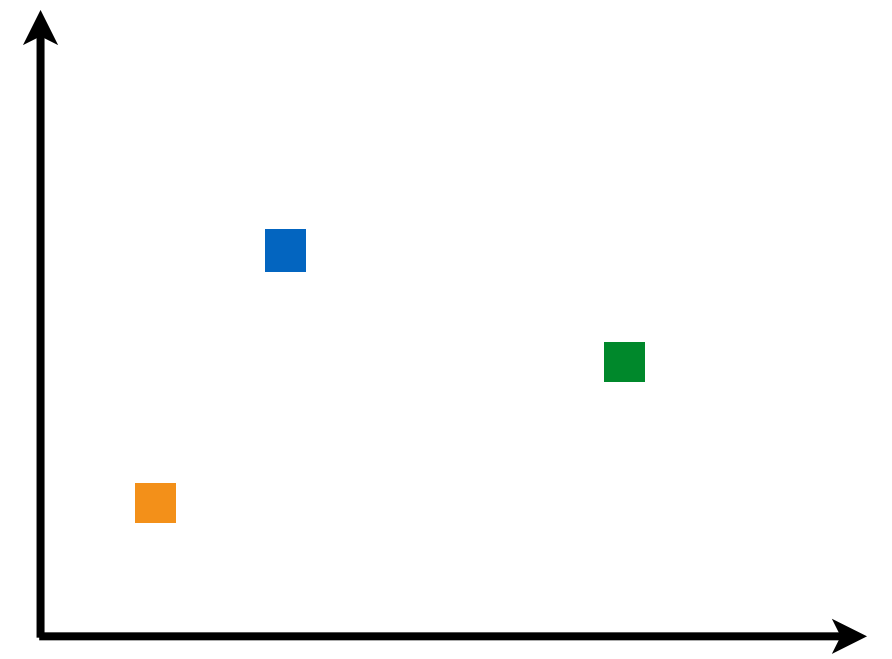
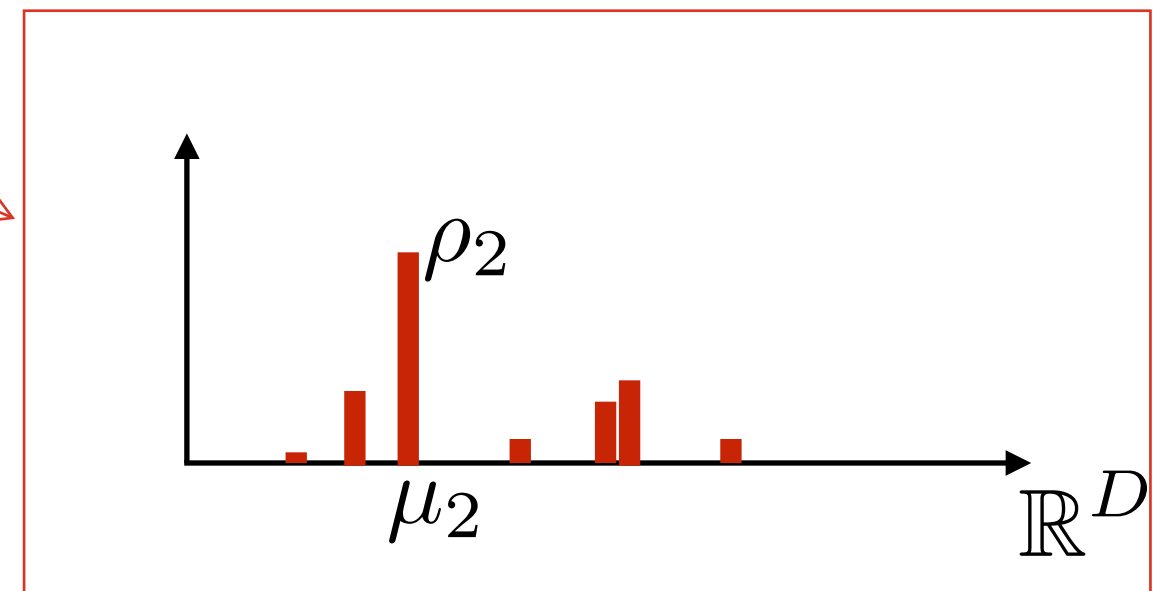
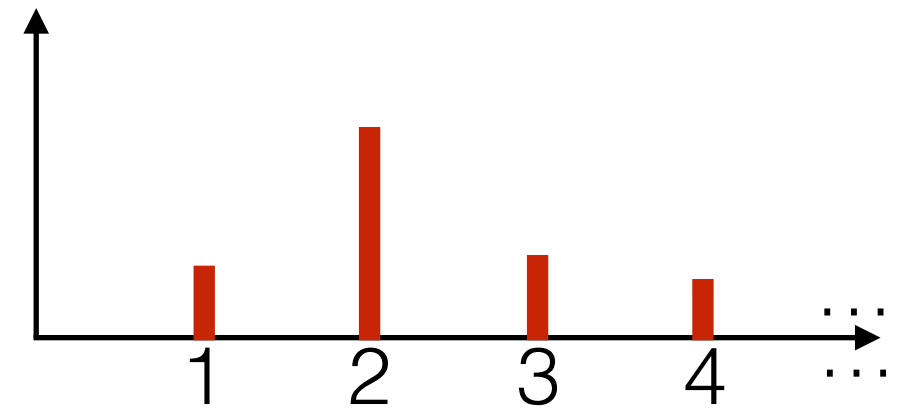
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- i.e. $G = \sum_{k=1}^{\infty} \rho_k \delta_{\mu_k}$



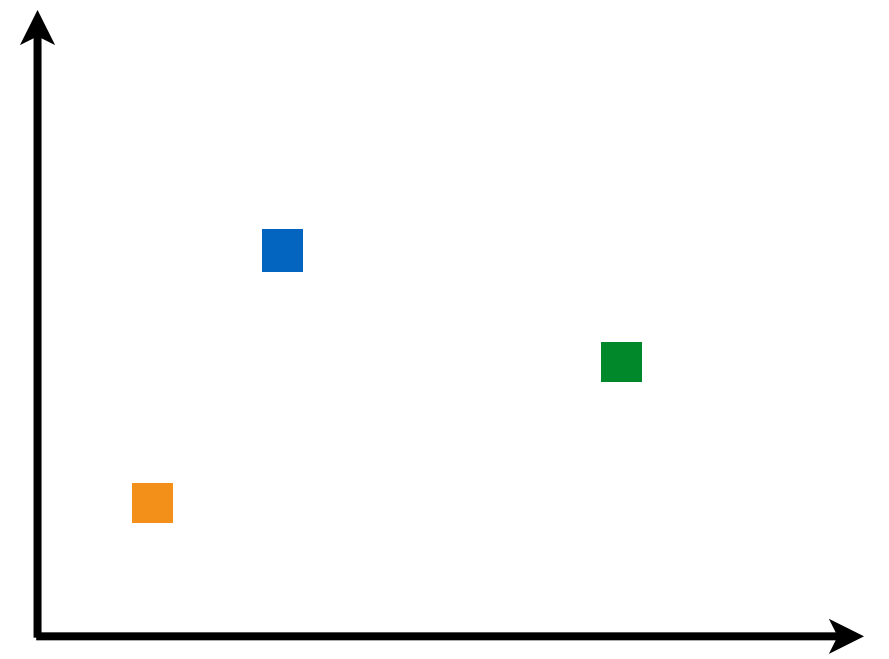
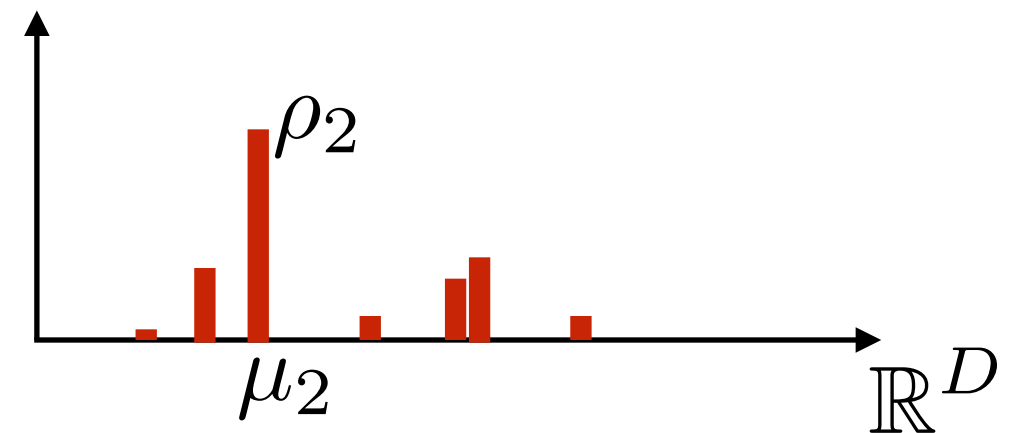
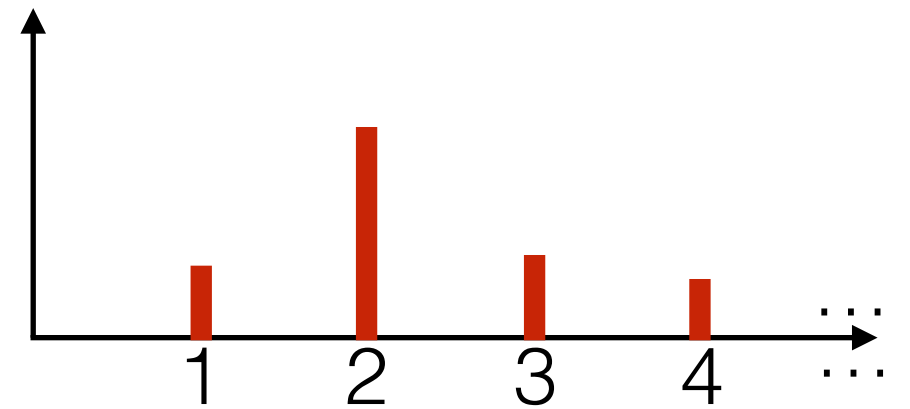
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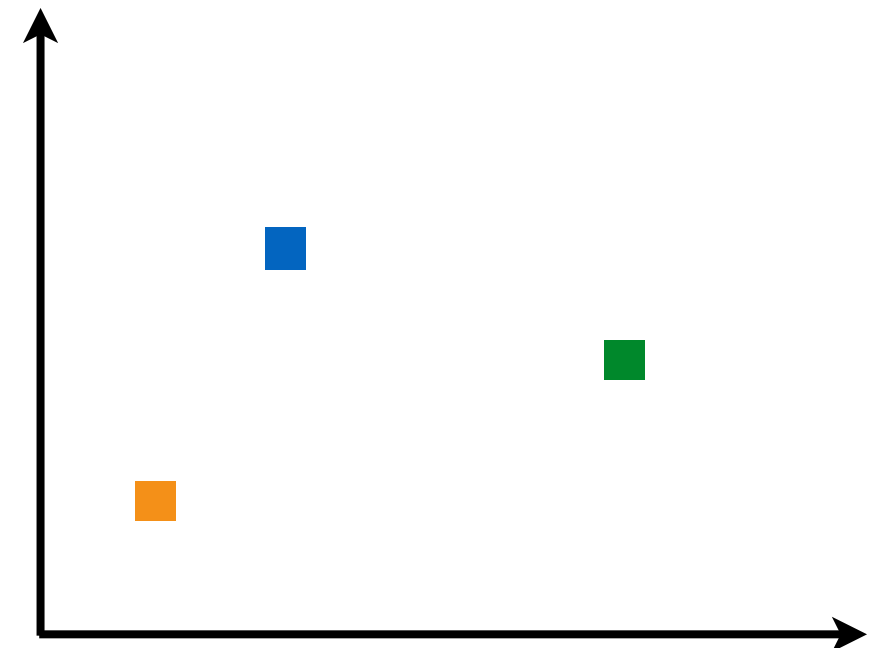
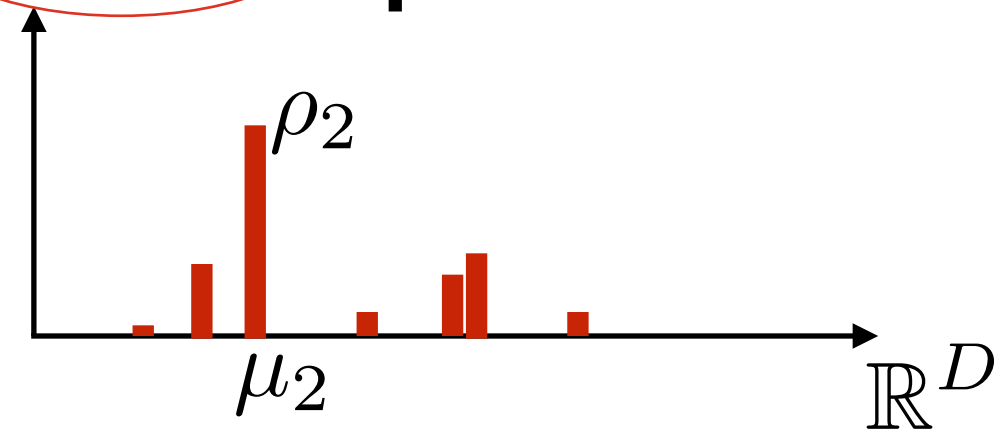
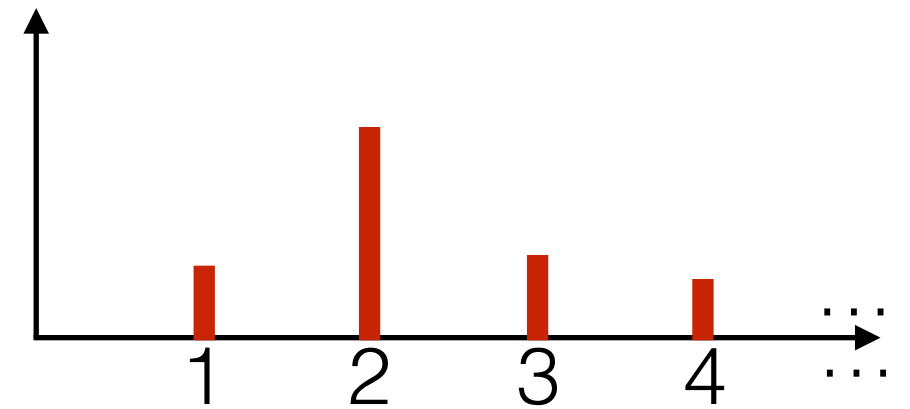
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- i.e. $G = \sum_{k=1}^{\infty} \rho_k \delta_{\mu_k} \stackrel{d}{=} \text{DP}(\alpha, \mathcal{N}(\mu_0, \Sigma_0))$ **Dirichlet process**

Base measure.
In general, this does not
have to be Gaussian.



Dirichlet process mixture model

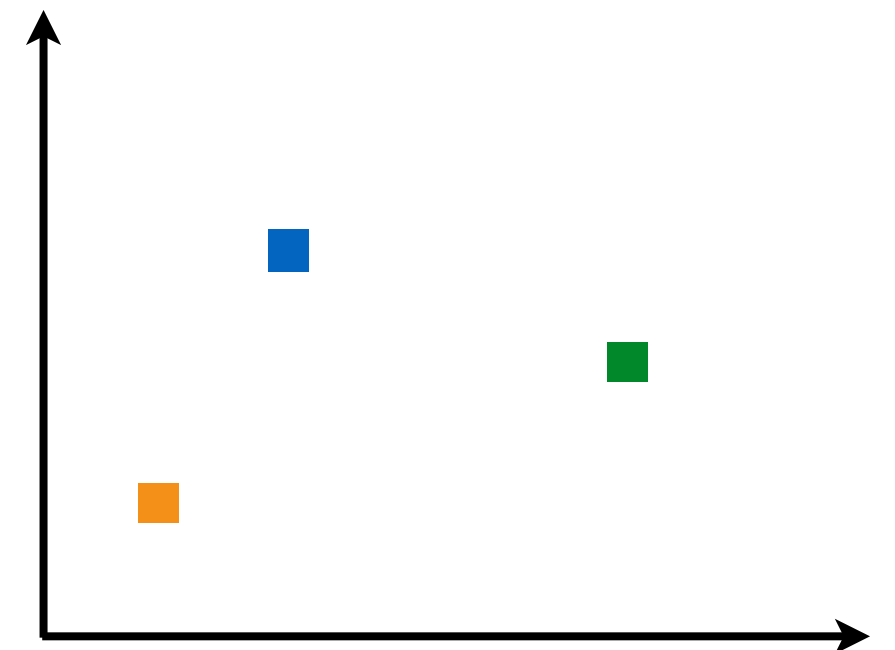
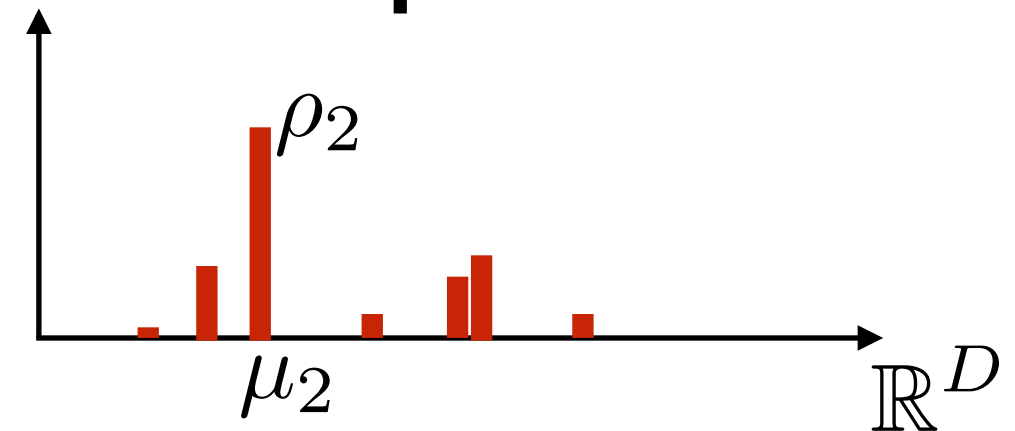
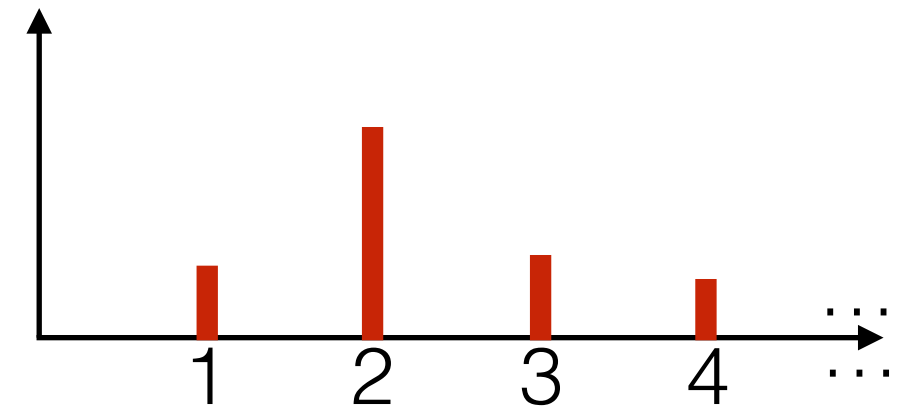
- Gaussian mixture model

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Dirichlet process



 $z_n \stackrel{iid}{\sim} \text{Categorical}(\rho)$

Dirichlet process mixture model

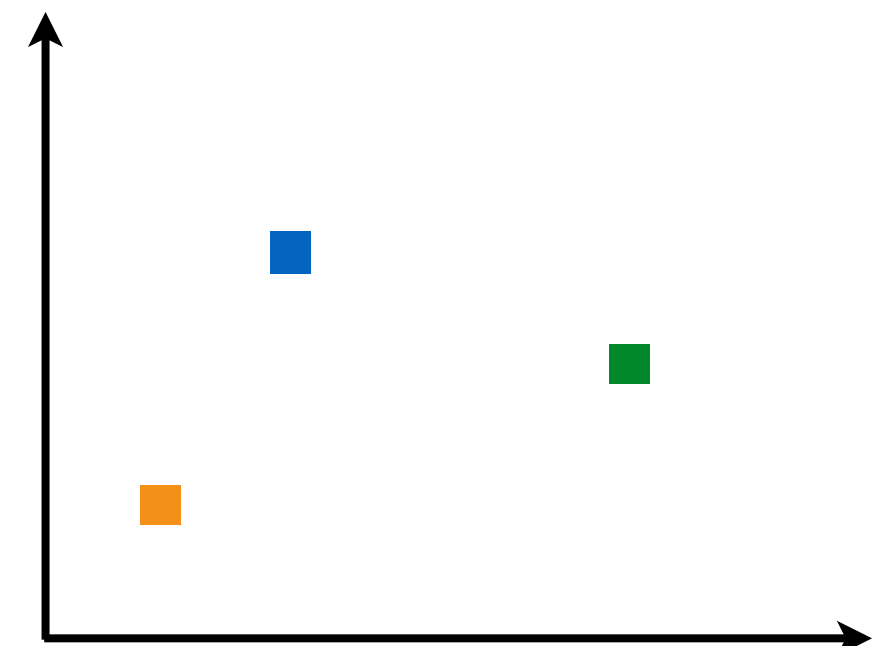
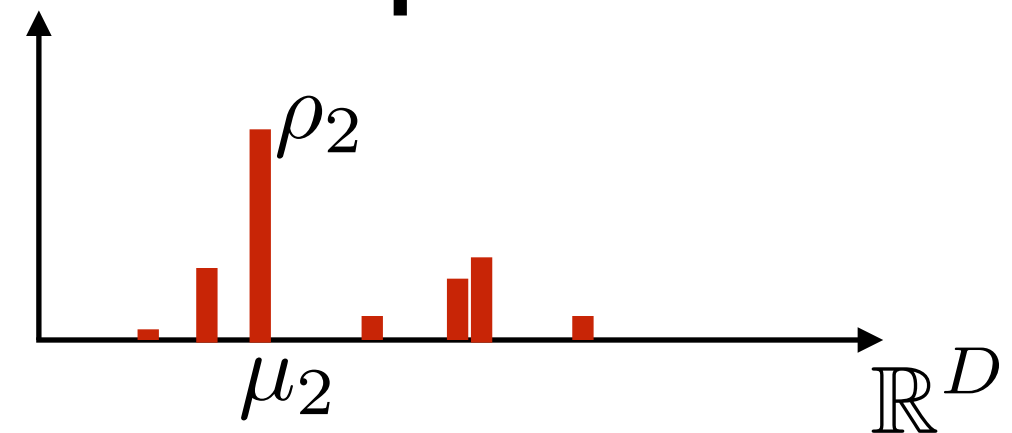
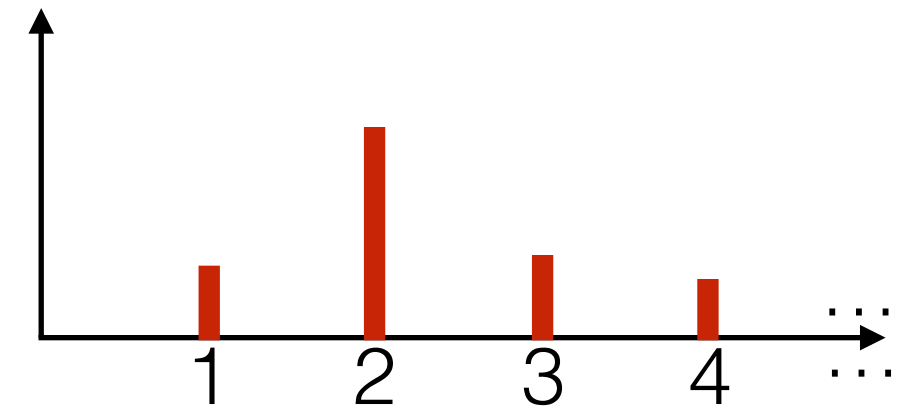
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Dirichlet process



$$z_n \stackrel{iid}{\sim} \text{Categorical}(\rho)$$

 $\mu_n^* = \mu_{z_n}$



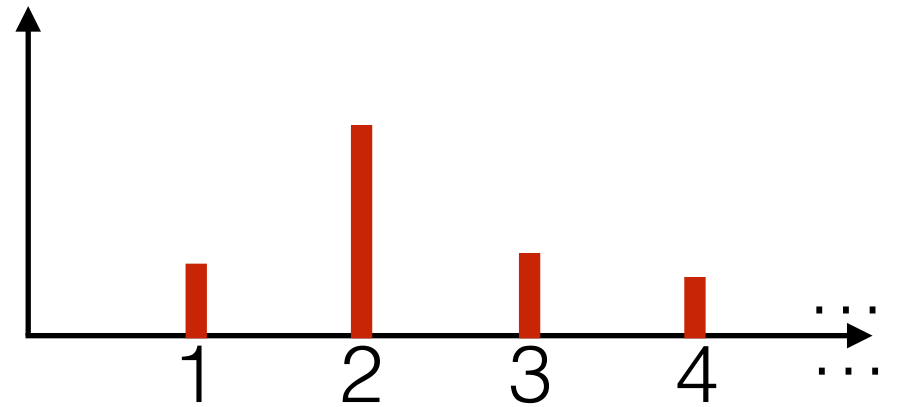
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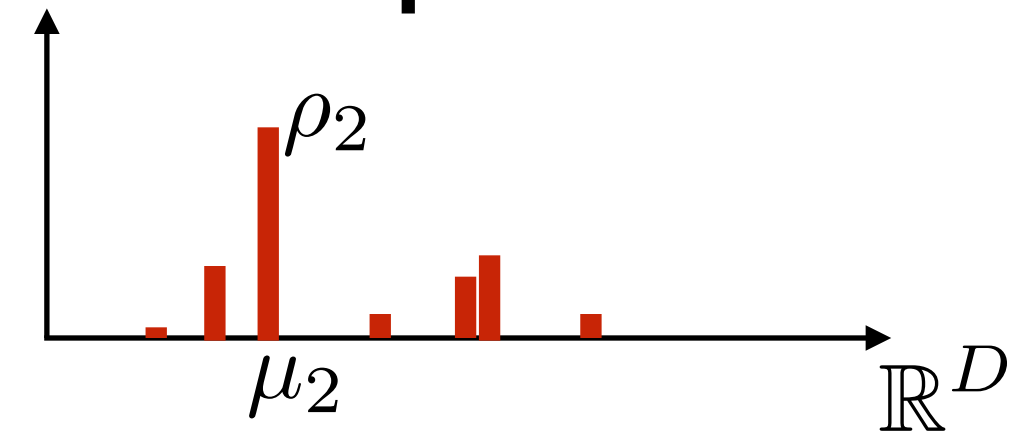
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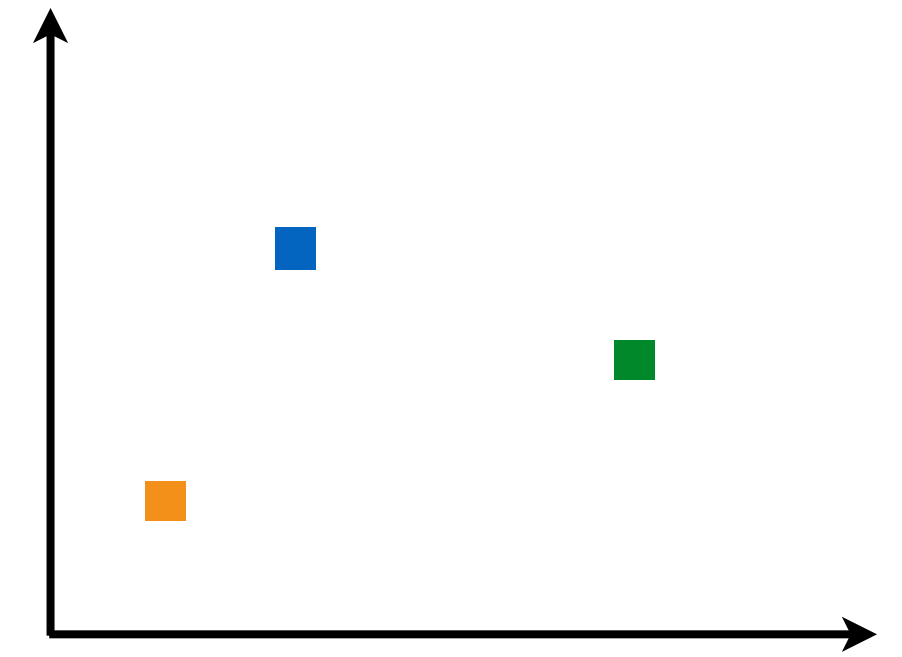


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Dirichlet process mixture model

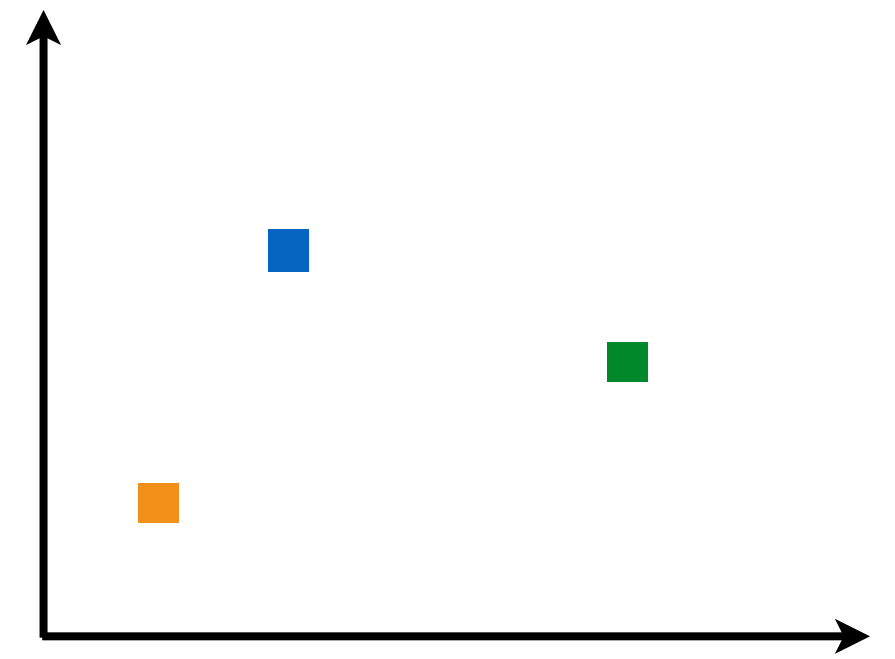
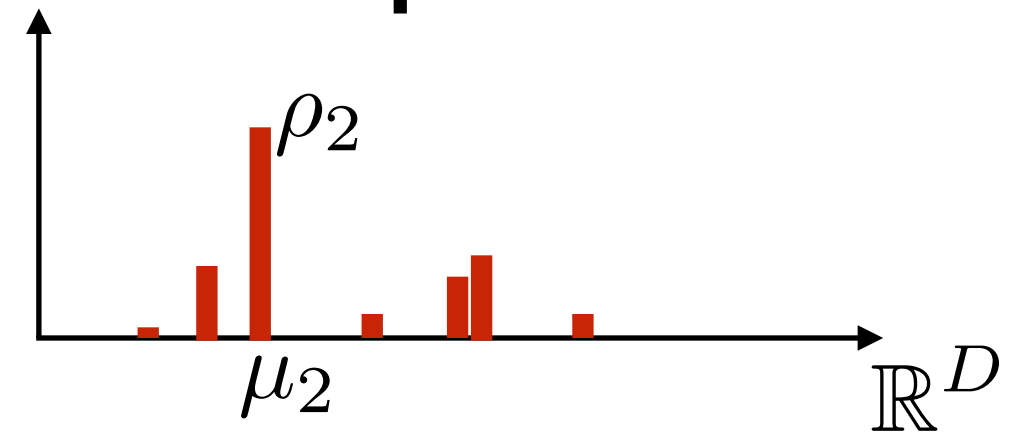
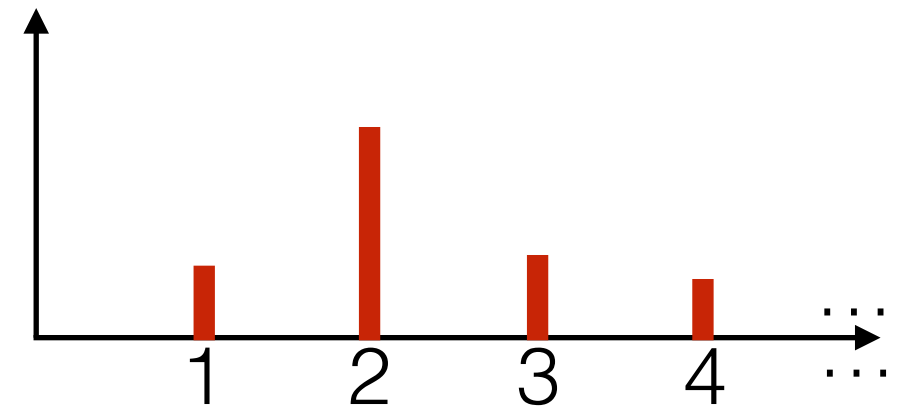
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Dirichlet process mixture model

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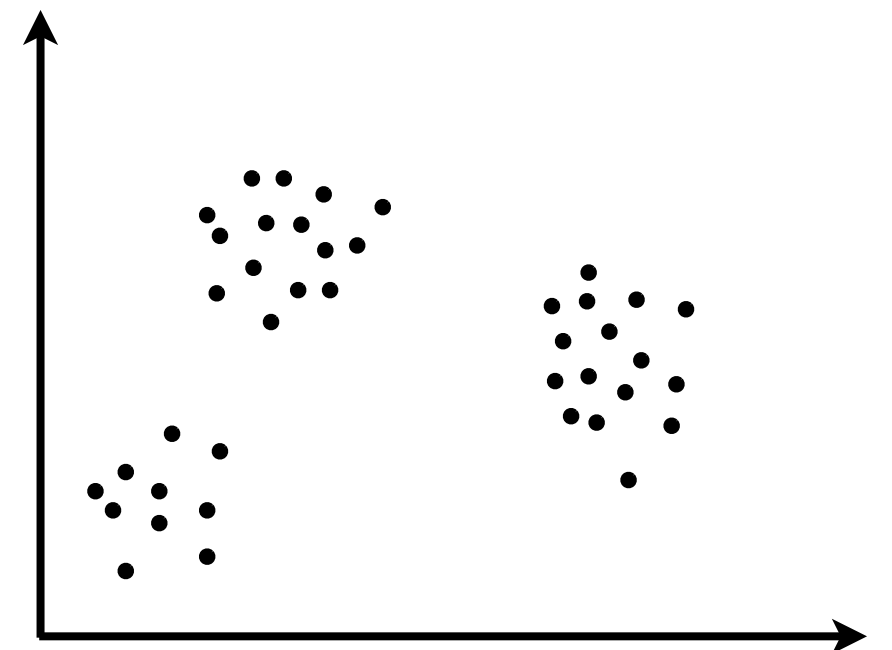
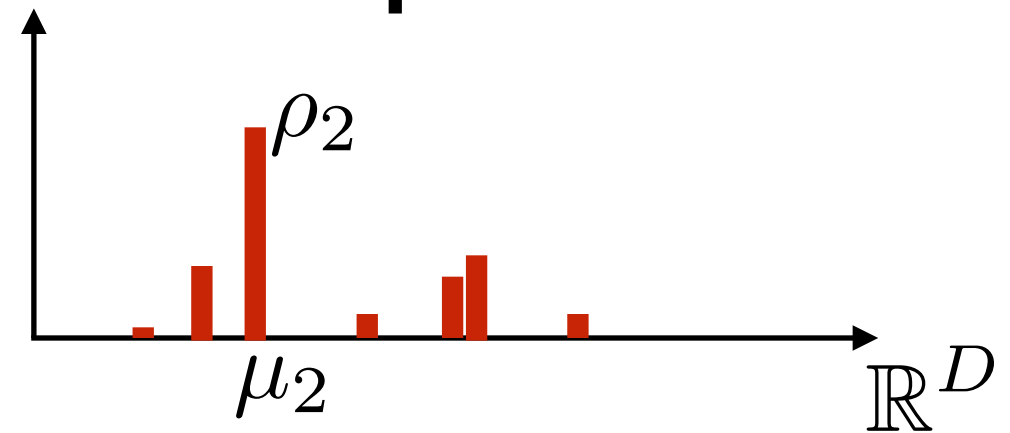
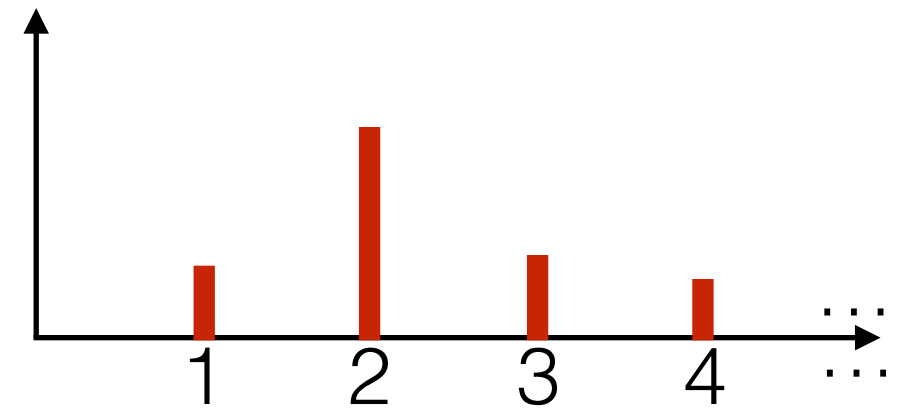
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Dirichlet process mixture model

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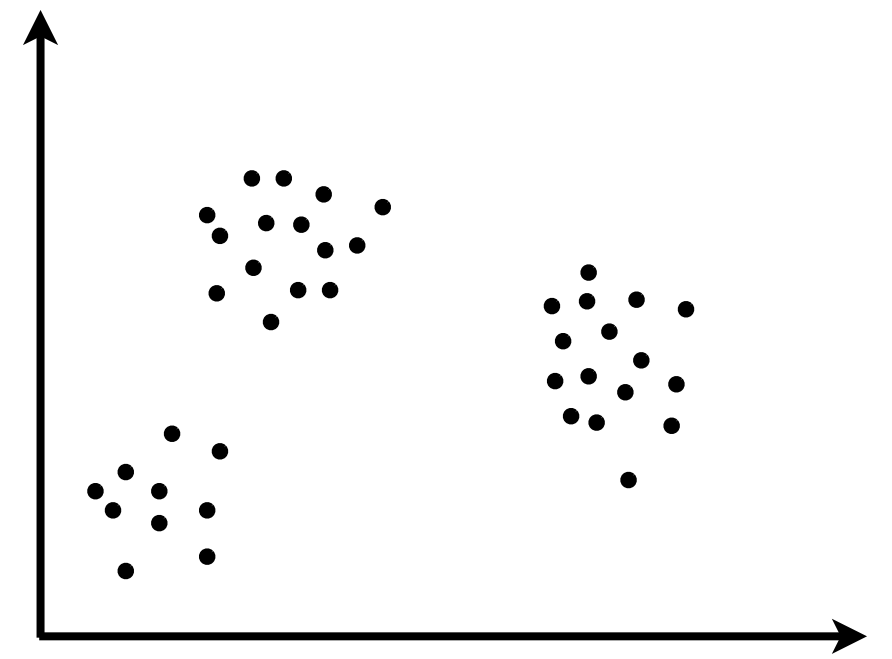
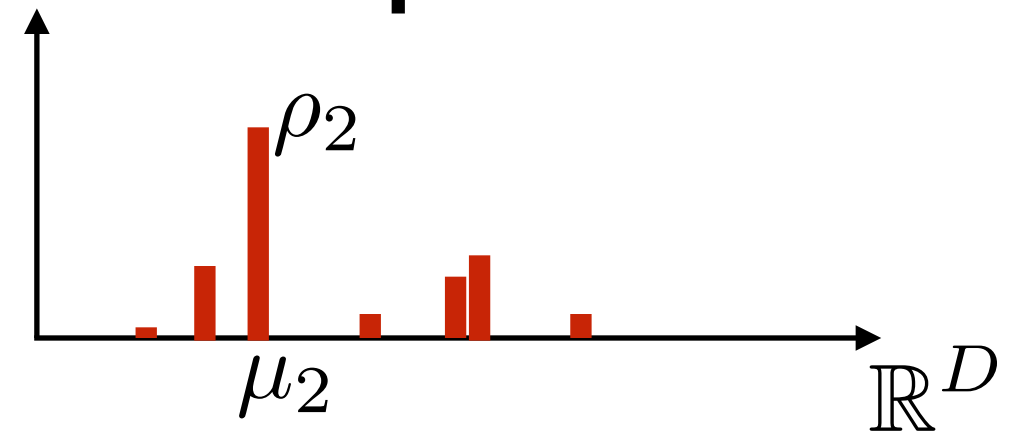
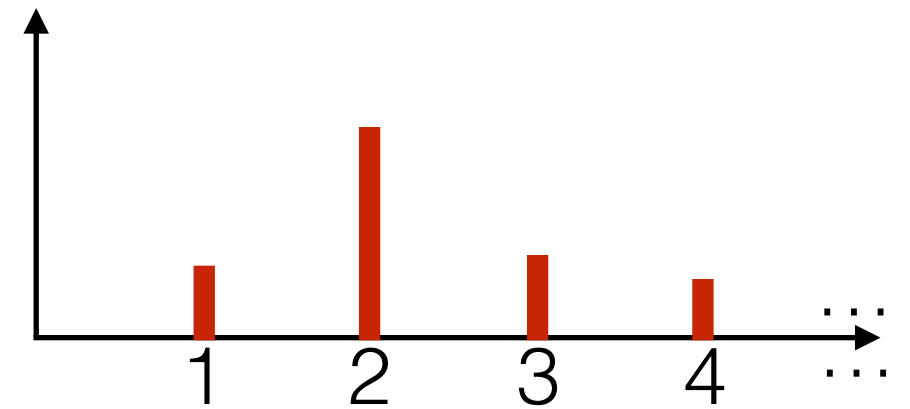
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[demo] 



Roadmap

- Example problem: clustering
- Example NPBayes model: Dirichlet process
- Big questions
 - Why NPBayes?
 - What does a growing/infinite number of parameters really mean (in NPBayes)?
 - Why is NPBayes challenging but practical?

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